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YIELD TOKENIZATION AND THE EMERGENCE OF CRYPTO-NATIVE FIXED INCOME MARKETS: EVIDENCE FROM PENDLE

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RESUMO

<TEXTO DO RESUMO EM PORTUGUÊS — parágrafo único, entre 150 e 500 palavras, sem citações e evitando siglas. Versão em português do Abstract.>

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ABSTRACT

Decentralized Finance (DeFi), an ecosystem of blockchain-based financial applications that operate without traditional intermediaries**,** has developed a wide range of mechanisms through which digital assets generate yield. For most of its history, however, that yield could not be traded on its own: it was simply the outcome of holding a position. This dissertation studies yield tokenization, the separation of a yield-bearing asset into a claim on the principal and a claim on the future yield, each tradable independently. Its central question is whether yield tokenization represents a step in the maturation of DeFi toward a possible crypto-native fixed income market. DeFi grew out of a technology-driven environment, and yield tokenization is examined as a point at which it begins to adopt the concepts and conventions of traditional fixed income. The Pendle protocol, the leading application of this idea, is examined as a case study through a qualitative and conceptual analysis based on the academic literature, the protocol's whitepapers and documentation, and its public code. The analysis develops an imperfect analogy between these tokenized instruments and traditional fixed income, including instruments bought at a discount and redeemed at maturity, rates of return implied by market prices, and maturities that together form something comparable to a yield curve. It also identifies the points at which the analogy stops holding, since the two sets of instruments exist in different environments, one consolidated by institutions over a long history, the other young, built in code, and still defining its rules. It then shows that supporting such instruments required more than creating the tokens: the protocol had to redesign the automated market maker through which they trade, so that time to maturity and rates of return, rather than only prices, are built into the trading mechanism. This adaptation brings on-chain trading closer to the conventions under which fixed income is traded. It further examines the risks specific to tokenized yield, including smart contract, liquidity, collateral, governance and regulatory risks. The findings suggest that yield tokenization should be understood not merely as a new class of financial instruments, but as **an early stage in the formation of** a crypto-native fixed income market structure. On this basis, the dissertation argues that Pendle is best understood not as a single instrument but as a market structure assembled inside a protocol, encompassing issuance, pricing, trading and the alignment of the tokens with the yield they represent. Within the protocol, this structure is organized in four layers: the instrument, pricing by yield, liquidity infrastructure, and the convergence and arbitrage mechanisms that keep rates aligned. Together, they **seek to** reproduce on-chain the functions that a fixed income market requires, from issuance to price formation. A further contribution lies in the approach itself.

Digital finance is a hybrid field, neither purely financial nor purely technological. It resists the familiar categories through which each side tries to read it, and most of the literature still examines it from one side only. By integrating concepts from traditional finance, blockchain infrastructure and DeFi protocol design into a single analytical framework, the work treats the two perspectives as inseparable.

Keywords: Decentralized Finance (DeFi), Yield Tokenization, Pendle Protocol, Fixed Income, PT/YT, On-Chain Yield, Crypto Markets, DeFi Interest Rates, Smart Contract Risk.

1 INTRODUCTION

A note on terminology. This dissertation sits between finance and technology, and many of its terms belong to one field but not to the other. A glossary at the end of the work defines each of them and, where the two fields read a term differently, gives both readings. The reader is invited to consult it whenever a term is unfamiliar.

1.1 Background and Historical Context

To understand the development of Decentralized Finance (DeFi), it is important to note that the crypto ecosystem did not emerge from financial-market actors. Its origins lie in the Cypherpunk Movement, a community of cryptographers, software developers, and open-source contributors concerned with privacy, decentralization, censorship resistance, and reduced dependence on centralized intermediaries. Bitcoin was introduced in this context as a peer-to-peer electronic cash system, not as a speculative financial instrument [19]. DeFi grew out of these same communities [1], and its continued evolution, through staking, liquid staking, restaking, and yield-bearing assets [7], produced increasingly sophisticated on-chain financial structures. Among these developments, yield tokenization is the object of this dissertation: protocols such as Pendle separate the principal component of a yield-bearing asset from its future yield stream, creating tradable instruments with maturity structures and implied rates [12], functionalities economically analogous to fixed income concepts traditionally observed in bond markets [8.2].

Throughout this work, these instruments are compared with traditional fixed income, and it is important that this comparison be read with a fundamental asymmetry in mind. Fixed income markets in their current form are the product of decades of institutional, legal, and technical development. Crypto-native fixed income is a few years old and still in formation. Holding one to the standard of the other would be structurally uneven, and this dissertation therefore treats the comparison as an analytical reference point, not as a destination. The term “crypto-native” is used in this sense: it designates a market of its own kind, built within the crypto environment. If DeFi develops structure, stability, and predictability of its own, these properties are likely to emerge from its own substrate: code, smart contracts, on-chain settlement, and transparency of protocol rules, rather than from replicating institutional forms designed for a different technological environment.

Pendle was chosen as the case study for four reasons. It is the leading protocol in yield tokenization. It is the only one that brings together, within a single protocol, the instrument, the mechanism that prices it and the infrastructure through which it trades, which is what this dissertation examines. It operates across several underlying assets and several maturities, which makes it possible to observe something comparable to a term structure. And its whitepapers, documentation and code are public, which allows the analysis to be verified.

1.2 Tokenization and Yield Tokenization in Two Readings: Technical Construction and Economic Meaning

This section introduces the two central terms through both lenses, delimits the scope of the work and clarifies what yield means in this context.

Many of the concepts in this dissertation carry two complementary readings. The technical reading describes how a concept is implemented in code and infrastructure. The economic reading describes what it means as a claim or financial instrument. Neither reading alone is sufficient, and both are kept in explicit dialogue throughout the analysis and in the glossary. This section introduces the two central terms through both lenses, delimits the scope of the work and clarifies what yield means in this context.

Tokenization is the process of creating a smart contract on a blockchain that implements a token standard (such as ERC-20) and maintains an on-chain ledger of balances associating token units with specific addresses. The token is a set of programmable rules and balances, not a digitized version of the underlying asset. Economically, tokenization defines the right that the token represents, such as ownership, credit, yield or another claim, and its relationship to the underlying asset, which may exist off-chain (real estate, fiat currency) or on-chain (native crypto-assets, yield-bearing positions).

Yield tokenization is a specific application of this process. A yield-bearing asset is decomposed into two separately tradable claims: the Principal Token (PT), redeemable for the underlying at maturity, and the Yield Token (YT), representing the future yield stream until maturity. Technically, this is implemented by depositing the underlying into a smart contract that mints the two new tokens, following the ERC-5115 standard specifically designed for wrapped yield-bearing assets [27]. Economically, this creates two derivative instruments whose combined content is equivalent to the original position, but whose independent tradability allows each to be priced and held separately.

In addition, this dissertation focuses exclusively on crypto-native fixed income, that is, fixed income that is on-chain in origin, in which cash flows are generated within decentralized protocols such as staking, lending, liquidity provision, and restaking. It excludes Real World Asset (RWA) tokenization, in which the underlying fixed income remains off-chain: pre-existing traditional instruments are brought on-chain for distribution rather than created from endogenous DeFi cash flows.

It is important to make explicit that yield itself has a different meaning in traditional fixed income and in DeFi. In traditional fixed income, yield is the return an investor receives for lending money: a borrower, such as a government or a company, promises to repay the principal with interest, and the yield reflects that promise and the risk that it will not be kept. In DeFi, there is usually no borrower making a promise. The return comes from the operation of the protocols themselves: a lending protocol collects interest from those who borrow against collateral, a decentralized exchange collects trading fees for the liquidity it provides, and a proof-of-stake network pays those who lock their tokens to help secure it. Each of these mechanisms produces a flow of yield, and it is this flow that yield tokenization separates from the principal and makes tradable. Chapter 2 examines these sources in detail.

1.3 Research Problem

DeFi protocols were built by technologists, and their early design reflected engineering concerns rather than financial ones. The central problem of this dissertation is whether these protocols are now incorporating the concepts of traditional finance, specifically fixed income, into their design, and whether the result is a market that combines both fields rather than belonging to either. Yield tokenization is the case through which this problem is examined, and Pendle is the protocol in which it can be observed most completely.

Existing studies describe the Principal Token (PT) and the Yield Token (YT) and stop there. Few works examine the apparatus that Pendle built around them inside the protocol: the automated market maker, the order book and the mechanisms that keep prices aligned, which together form a more complete fixed income market. Less attention has also been devoted to where the analogy with traditional fixed income holds and where it breaks down. This dissertation addresses both gaps.

This leads to the central research question of this dissertation:

Does yield tokenization represent a step in the maturation of DeFi toward a crypto-native fixed income market?

1.4 Research Objectives

To answer this question, the objective of this research is to examine how tokenized yield structures replicate, reformulate or diverge from traditional fixed income concepts, using Pendle as the central case study.

Specific Objectives

1. Analyze the operational and economic structure of Pendle, including how the protocol separates yield-bearing assets into Principal Tokens (PTs) and Yield Tokens (YTs), the role of Standardized Yield (SY) assets, and the architecture of Pendle's Automated Market Maker (AMM).
2. Examine the financial logic underlying PTs and YTs in relation to traditional fixed income concepts (including zero-coupon bonds, discounting mechanisms, term structures, and fixed-yield exposure), distinguishing analogies that hold rigorously from those that break down. In particular, this dissertation evaluates the conditions under which the PT/zero-coupon bond analogy is strict, namely when the underlying is a stablecoin-denominated yield-bearing asset, versus cases where the underlying is a volatile asset (for example, stETH), in which the redemption value at maturity is itself subject to price variation.
3. Study implied yields and maturity-based pricing mechanisms, investigating how implied interest rates emerge from PT pricing and how maturity structures within Pendle resemble certain characteristics of traditional fixed income markets.
4. Evaluate the layered risk structure embedded in tokenized yield markets, identifying the multiple sources of risk incorporated into DeFi yields, including smart contract risk, liquidity risk, depeg risk, protocol dependency, staking and restaking exposure, incentive emissions, and composability-related vulnerabilities.

1.5 Research Contribution

This dissertation does not aim to develop a new mathematical pricing model or propose a novel quantitative framework for valuing DeFi yield instruments. Its primary contribution lies instead in providing a conceptual and financial interpretation of yield tokenization as an emerging form of crypto-native fixed income infrastructure.

Finally, the dissertation integrates concepts from traditional finance, blockchain infrastructure, and DeFi protocol design into a unified analytical framework, seeking to bridge the gap between financial theory and emerging decentralized market architectures. This integration is itself a contribution, because the two bodies of literature have developed largely in isolation from one another. Work on DeFi has been shaped primarily by contributors from software engineering and cryptography, whose analytical approach emphasizes protocol mechanics and cryptographic primitives but often does not engage with the mature literature on financial intermediation, credit, and market structure. Work on traditional finance, in turn, has treated DeFi predominantly as a source of risk or speculation, sometimes without engaging deeply enough with the technical constructions that make the ecosystem operate. This dissertation attempts to occupy the space between the two, applying financial theory to constructs whose mechanics originate in on-chain smart contracts, and applying the analytical tools developed for DeFi to identify the specific technical properties that give those constructs their economic meaning. In doing so, it offers a foundation for future research on crypto-native fixed income systems and programmable financial infrastructure, research that will require the same dual engagement, addressing an audience neither exclusively financial nor exclusively technological.

The technological reading matters for a specific reason. In traditional finance, a professional does not need to know how the systems of a clearing house or a custodian operate, because an institution stands behind them and guarantees the result. The technology is hidden behind a name that the market trusts. In DeFi there is no such name. The code performs the functions that the clearing house, the custodian and the contract perform in traditional markets. Understanding the mechanism is therefore not a matter of understanding a product, but of understanding who the counterparty is, how settlement takes place and where the risk lies. This is why a reader trained in finance needs to understand the technology to an intermediate level, and why this dissertation presents each mechanism through both readings. The automated market maker is the clearest case: it is not a piece of infrastructure in the background, but the

market maker itself, and no market can be evaluated without knowing how it forms prices and provides liquidity. The structure of the AMM is examined in detail in Chapters 3 and 4.

1.6 Methodology and Dissertation Structure

This dissertation adopts a qualitative case-study strategy, in which Pendle Finance is examined as a representative instance of yield tokenization in DeFi. The analysis combines three complementary elements: (i) a literature review that connects DeFi, fixed income theory, and the emerging literature on yield tokenization; (ii) a detailed case study of the Pendle protocol, based on official documentation, whitepapers, and technical analyses; and (iii) selective empirical evidence drawn from public on-chain data sources such as DefiLlama, Dune Analytics, and Etherscan. The empirical component is not econometric; its purpose is to provide market-oriented context and to illustrate the economic relevance of the mechanisms discussed conceptually.

Pendle was chosen as the case study for four reasons. It is the dominant protocol in yield tokenization, holding most of the market's value. It is the only one that brings together, within a single protocol, the instrument, the mechanism that prices it and the infrastructure through which it trades, which is precisely what the research question requires. It operates across several underlying assets and several maturities, which makes it possible to observe something comparable to a term structure. And its whitepapers, documentation and code are public, which allows the analysis to be verified.

The remainder of the dissertation is organized as follows:

Chapter 2 reviews the relevant literature and theoretical framework, covering DeFi yield generation, traditional fixed income concepts, and the emerging literature on yield tokenization and implied rates in crypto markets.

Chapter 3 examines the main sources of yield generation in DeFi, including lending, liquidity provision, staking, restaking, and composability.

Chapter 4 constitutes the core of the dissertation and analyzes yield tokenization through the case of Pendle, with particular attention to PT, YT, SY, implied rates, and market design.

Chapter 5 discusses the main risks, structural limitations, and regulatory challenges associated with tokenized yield markets.

Chapter 6 presents an empirical and market-oriented analysis based on protocol data, liquidity, maturity structures, and on-chain yield dynamics.

Chapter 7 develops the broader discussion of the dissertation's main findings. It discusses the extent to which yield tokenization represents a maturation process within DeFi, evaluating whether protocols such as Pendle indicate the emergence of more professionalized and financially structured crypto-native markets.

Chapter 8 concludes.

2 LITERATURE REVIEW AND THEORETICAL FRAMEWORK

This chapter establishes the theoretical and academic foundation for analyzing yield tokenization as a potential form of crypto-native fixed income infrastructure. The discussion is organized into three main areas. First, it reviews the literature on DeFi and yield generation mechanisms. Second, it presents the fixed income concepts required to interpret tokenized yield structures. Third, it discusses the emerging literature specific to yield tokenization, implied interest rates, and on-chain yield markets. A final section identifies the gap that this dissertation addresses.

The purpose of this chapter is not to provide an exhaustive review of the entire DeFi ecosystem, but rather to identify the concepts most relevant to understanding how protocols such as Pendle may transform future yield streams into tradable financial instruments.

2.1 DeFi and Yield Generation

Decentralized Finance (DeFi) refers to a set of blockchain-based financial protocols that provide services such as trading, lending, borrowing, liquidity provision, and asset management without relying on traditional intermediaries. These protocols operate through smart contracts, allowing financial interactions to be executed programmatically and transparently on public blockchains [1], [2].

The early development of DeFi was strongly technology-driven. Protocols such as Uniswap, Compound, Aave, and MakerDAO created basic building blocks for decentralized exchange, lending, collateralized borrowing, and stablecoin issuance. These applications demonstrated that financial functions traditionally provided by banks, brokers, or exchanges could be replicated through programmable infrastructure operating on public blockchains [4].

A central feature of DeFi is the generation of yield. In traditional finance, yield usually arises from lending, credit risk, duration risk, dividends, or ownership of income-producing assets. In DeFi, yield can arise from several distinct mechanisms, including lending interest, liquidity provision fees, staking rewards, protocol incentives, governance token emissions [18] and automated vault strategies.

The literature on DeFi yield commonly distinguishes between different sources of return. Lending protocols, such as Aave and Compound, generate yield from borrower interest [5], [21]. Automated Market Makers (AMMs) [16], such as Uniswap [23], [24], [25] and Curve,

generate yield from trading fees paid to liquidity providers. Staking mechanisms generate rewards for securing proof-of-stake networks: participants lock their tokens to help validate transactions, and the network pays them for this service. Vault strategies, such as Yearn, aggregate multiple yield sources into single deposit interfaces.

This literature is useful for understanding the evolution of DeFi from simple yield generation toward more complex financial structures. However, many studies on DeFi yield focus on individual strategies, protocol mechanics, or portfolio diversification. They often treat yield as an outcome to be compared across protocols, rather than as a financial object that can itself be separated, priced, and traded.

This distinction is important for this dissertation. Yield tokenization does not merely generate yield; it transforms expected future yield into tradable instruments. Therefore, it represents a more advanced stage in the development of DeFi markets.

2.2 Fixed Income Theory

To analyze yield tokenization, it is necessary to introduce basic concepts from fixed income theory. Fixed income markets are built around instruments that promise future cash flows, usually with defined payment dates, maturities, and discounting mechanisms. Bonds, zero-coupon bonds, yield curves, and term structures are central to this framework.

A zero-coupon bond is particularly relevant for this dissertation. Unlike a coupon-paying bond, a zero-coupon bond does not make periodic interest payments. Instead, it is issued at a discount and redeemed at face value at maturity. The investor's return is determined by the difference between the purchase price and the redemption value.

Formally, the price of a zero-coupon bond is the discounted value of its redemption amount:

$$P = \frac{F}{(1+r)^T},$$

where P is the current price, F the face value redeemed at maturity, r the annualized yield, and T the time to maturity in years. Solving this relationship for r gives the yield implied by the price, the same operation through which, in Section 4.3, an implied rate is extracted from the market price of a PT.

This logic is important because Principal Tokens (PTs) in Pendle exhibit a similar economic structure. A PT can be purchased at a discount and redeemed for the underlying asset

at maturity. This creates an implied yield, derived from the relationship between the current PT price, the redemption value, and the time remaining until maturity.

A second concept follows from the first: a coupon-paying bond can be decomposed into separately tradable pieces. In the U.S. Treasury STRIPS market, the principal and each coupon payment are traded as independent zero-coupon claims [36]. This decomposition, one instrument split into a principal claim and a stream of future payments, is the closest traditional antecedent of the PT/YT separation examined in Chapter 4.

One further set of instruments matters for this dissertation: interest rate derivatives. The most common of them, the interest rate swap, is a contract in which two counterparties exchange payment streams on an agreed notional amount, one fixed and one floating, without either party holding the underlying asset [35]. Its value derives from the path of the rate itself, and each payment depends on the counterparty honouring the contract. This structure is the reference against which Chapter 4 examines the Yield Token.

A zero-coupon bond also has the simplest duration of any fixed income instrument: its duration equals its time to maturity, a property that returns in the analysis of PT pricing and of the Pendle AMM in Chapter 4.

However, this analogy must be treated carefully. Traditional fixed income instruments are usually embedded within regulated financial systems, with legal claims, issuer obligations, credit ratings, settlement institutions, and benchmark yield curves. By contrast, PTs are smart-contract-based claims within DeFi protocols. Their implied yields reflect not only time value, but also protocol-specific risks, technological dependencies, and the composability of underlying yield sources. The analogy is strongest when the underlying yield-bearing asset is stablecoin-denominated and the redemption value is therefore predictable in fiat terms; when the underlying is a volatile asset, the redemption value itself is subject to price variation and the analogy weakens.

Another important concept is the yield curve. In traditional finance, there is not one yield curve but several: the curve built from government bonds serves as the benchmark, and separate curves exist for corporate bonds of each credit rating, such as AAA or AA, each lying above the benchmark by a spread that reflects its credit risk. The government curve is commonly used as the reference for discounting, valuation, risk management, and monetary policy analysis. In DeFi, there is no universally accepted risk-free curve [10]. Instead, implied

yields may vary across protocols, assets, chains, maturities, and liquidity conditions, and cannot be interpreted as risk-free rates in the traditional sense.

The structure of several curves, one for each class of risk, has a counterpart in yield tokenization. In Pendle, each underlying asset has its own market, with its own maturities and its own implied rates, so that staked ETH, a yield-bearing stablecoin and a lending position each produce a separate curve. The resemblance stops at the structure, however. In traditional markets, every bond on the AAA or the AA curve promises the same currency, and the distance between the two curves is credit risk alone. In Pendle, the distance between two curves reflects three things at once: the expected yield of the underlying asset, which differs between staking and a stablecoin strategy, the risk of the protocol that generates it, and the currency in which the asset is denominated. A gap between two Pendle curves is therefore not a pure credit spread, and cannot be read as one.

This raises a question that a reader trained in finance will ask at once: if a DeFi position carried the same risk as a bond of a given rating, should arbitrage not bring the two yield curves together? The question is a fair one, and the answer depends on three conditions, concerning the nature of the risk, the mobility of capital and the source of the payment. Section 4.3 examines each of them and explains why the two curves do not converge today.

Therefore, the fixed income framework provides useful concepts, but it cannot be applied mechanically. The central challenge is to understand where the analogy between traditional fixed income and yield tokenization is meaningful, and where it breaks down.

2.3 Yield Tokenization Literature

Yield tokenization is a more recent development within DeFi [13]. It refers to the separation of a yield-bearing asset into two components: a principal component and a yield component. In Pendle, these are represented by Principal Tokens (PTs) and Yield Tokens (YTs).

The PT represents a claim on the underlying asset at maturity. The YT represents exposure to the future yield generated by the underlying asset until maturity. This separation allows users to obtain fixed yield, speculate on future yield movements, hedge yield exposure, or trade yield independently from the underlying principal.

Recent academic and technical literature has begun to explore these mechanisms. Work on crypto interest rates and yield curves highlights the difficulty of constructing benchmark

rates in a market without traditional bonds. In a bondless market, implied rates must often be inferred from derivatives, lending protocols, or other market instruments. Yield tokenization offers another possible source of implied rate information [10].

Another relevant line of research studies the pricing and hedging of tokenized yield instruments. These works are closer to the mathematical and technical modeling of yield tokenization. They examine how principal and yield components can be priced and how fixed-rate exposure can be created in DeFi [13], [8.2].

A closely related literature concerns the market infrastructure through which these instruments trade. Fixed-rate instruments cannot be priced by the formulas designed for spot markets, because the value of the instrument changes as maturity approaches even when no trade occurs. YieldSpace addressed this by introducing an invariant in which the pool quotes a constant interest rate rather than a constant price, taking time to maturity as an explicit input [28]. Notional Finance proposed an alternative construction, later adopted as the baseline for the Pendle V2 AMM [29], [17]. More recent work generalizes these designs to accommodate several maturities within a single pool [30], [31], while the microeconomic properties of constant function market makers have been formalized separately [32], [16]. This literature matters here because the implied rate observed in a yield tokenization protocol is produced by the market where the instruments trade, and not by the instrument alone.

A further strand of the discussion concerns how these instruments should be classified. The Yield Token gives its holder exposure to a rate rather than to an asset, which places it closer to interest rate derivatives than to the spot instruments usually examined in the DeFi literature. The protocol documentation itself describes YT positions as long or short exposure to yield [34], and the standard definitions of derivative instruments (underlying, notional amount, forward commitment, replication) provide the vocabulary for examining that description [35]. The academic literature on yield tokenization has so far concentrated on pricing and hedging rather than on classification, which leaves room for the analysis developed in Chapter 4.

The Pendle documentation and whitepapers are also important sources because they describe the protocol architecture, including the role of PTs, YTs, Standardized Yield (SY), and the Pendle AMM [11], [12], [17]. The SY token standard itself is defined outside the protocol documentation, as an Ethereum standard [27]. Although these materials are not academic articles in the traditional sense, they are necessary to understand the protocol's financial mechanics.

For this dissertation, the relevant contribution is not to develop a new pricing model. Instead, the aim is to interpret the existing literature and protocol evidence into a conceptual framework that interprets yield tokenization as a step in the maturation of DeFi toward a crypto-native fixed income market.

2.4 Literature Gap

Most DeFi surveys focus on how yield is generated, but less on how yield can be separated from principal, tokenized, priced, and traded as an independent financial object. Traditional fixed income theory offers useful analogies, but it does not fully account for the composability, smart contract risk, protocol dependency, and layered yield structures observed in DeFi. Technical papers on Pendle and related protocols provide detailed mechanics and pricing analyses, but do not always situate these instruments within the broader interpretive question of whether they represent an emerging fixed income market.

Two aspects of this intersection remain underexplored. The first is that the literature examines the separation of yield from principal, but not the fact that Pendle delivers the complete market structure required for a fixed income market to function: the instrument, the mechanism that prices it, and the infrastructure through which it trades, operating together. The second is the systematic relation between traditional fixed income and these on-chain constructions: which analogies hold, under which conditions, and where they break down has itself received little direct attention. This dissertation positions itself in these two gaps, taking Pendle as the case in which both can be examined at once.

The literature reviewed in this chapter suggests that yield tokenization may represent more than the creation of new financial instruments. Whether these innovations amount to the emergence of a crypto-native fixed income market, however, remains an open question. The following chapter therefore examines the mechanisms through which yield is generated in DeFi, since this yield is the raw material that yield tokenization transforms into tradable claims. Chapter 4 then develops the analysis of Pendle on that basis.

3 YIELD GENERATION IN DEFI

This chapter examines the main mechanisms through which yield is generated within Decentralized Finance (DeFi). The objective is to provide the conceptual foundation necessary to understand the emergence of yield tokenization and the development of more sophisticated on-chain financial structures.

Unlike traditional finance, where yield is generally associated with lending, credit exposure, or ownership of productive assets, DeFi allows yield to emerge from programmable interactions between multiple protocols, smart contracts, and incentive systems. This creates an ecosystem in which returns may originate simultaneously from lending activity, liquidity provision through AMMs, staking rewards, governance incentives, and protocol emissions.

Each mechanism in this chapter is presented through two readings, one from technology and one from finance: first, how the mechanism operates, and second, what type of return it generates and for whom. The purpose is not to survey DeFi protocols but to identify, in each mechanism, the type of yield it generates and the yield-bearing asset that results from it, since these are the assets whose yield tokenization will later separate from the principal. Understanding these mechanisms is therefore essential for analyzing, in Chapter 4, how Pendle transforms future yield streams into tradable financial instruments.

Sections 3.1 to 3.3 examine the three main sources in turn, lending, liquidity provision through AMMs, and staking, and Section 3.4 shows how yields from different sources combine, with incentives and emissions appearing within each of them.

3.1 Lending Protocols

Lending protocols represent one of the foundational layers of the DeFi ecosystem. Platforms such as Aave, Compound, and Morpho enable users to lend and borrow digital assets without relying on traditional banking intermediaries. Instead of centralized credit analysis, these systems operate through smart contracts and overcollateralized positions [5].

In these protocols, lenders supply assets to liquidity pools and earn yield from borrower interest payments. Borrowers, in turn, must provide collateral whose value exceeds the borrowed amount, reducing credit risk for the protocol.

Unlike traditional banking systems, DeFi lending markets typically operate with floating interest rates determined algorithmically by supply and demand conditions. The key variable is the utilization rate, that is, the share of the funds deposited in the protocol that is currently lent out. When utilization rises, borrowing costs tend to rise in order to attract

additional liquidity. Conversely, excess liquidity, meaning a low utilization rate, generally leads to lower lending rates.

These lending protocols introduced one of the earliest forms of on-chain yield generation and became central to the expansion of DeFi during the early stages of the ecosystem. However, the yield they produce remains structurally floating: users receive a variable rate that shifts continuously with pool conditions, without any built-in mechanism to lock in a fixed return over a defined horizon. This limitation, inherent to the pool-based lending design, is precisely what motivated the emergence of yield tokenization protocols such as Pendle, which allow the same underlying yield-bearing position to be separated into a fixed-return principal component and a variable yield component, as examined in Chapter 4.

3.2 Automated Market Makers (AMMs) and Liquidity Provision

Automated Market Makers (AMMs) represent one of the core components of the DeFi ecosystem. In contrast to traditional order-book exchanges, where trades are executed by matching buyers and sellers, AMMs use a peer-to-pool mechanism, in which users trade against liquidity pools, and asset prices are determined algorithmically through a conservation function, a mathematical rule that defines how the quantities of assets in the pool must adjust when trades occur [16]. Protocols such as Uniswap and Curve became central examples of this market structure.

In a simple AMM, liquidity providers deposit assets into a pool. Traders then exchange one asset for another by interacting with that pool, rather than with a specific counterparty. The pool therefore acts as the immediate source of liquidity for the trade. The pricing function updates the exchange rate according to changes in the pool's reserves. For example, in a pool containing ETH and USDC, buying ETH with USDC reduces the amount of ETH in the pool and increases the amount of USDC, changing the relative price implied by the pool.

The AMM is neither a fixed income market nor a currency market. It is part of the market environment: the mechanism through which DeFi provides liquidity in the absence of a market maker. In traditional markets, a dealer stands ready to buy and sell, and that is what makes an instrument liquid. In DeFi there is no dealer. Liquidity comes from a pool of assets deposited by participants, and a formula sets the price at which anyone can trade against that pool. The example that follows, a trader exchanging USDC for ETH, is a spot transaction, closer to buying an asset for dollars than to fixed income. It is chosen because it is the simplest case

in which the mechanism can be seen at work. The same mechanism is what gives liquidity to the Principal Token examined in Chapter 4, after the adaptations that a fixed income instrument requires. Providing liquidity to a pool is also itself a source of yield, since liquidity providers receive a share of the fees paid by traders.

The most widely used rule for how an AMM sets prices is the constant-product model used by Uniswap V2, represented by the formula $x \times y = k$, where x and y are the quantities of the two assets in the pool and k is the invariant preserved during swaps, that is, the number that must remain the same before and after each trade. This mechanism allows continuous liquidity, but it also means that larger trades move the pool further along its pricing curve and create greater price impact.

Suppose a pool initially contains 10 ETH and 20,000 USDC. The constant product is therefore $10 \times 20{,}000 = 200{,}000$. The initial price implied by the pool is 2,000 USDC per ETH. If a trader adds 2,000 USDC to buy ETH, the USDC reserve increases to 22,000. To preserve k=200,000, the ETH reserve must fall to approximately 9.09 ETH. The trader therefore receives about 0.91 ETH. Instead of receiving 1 ETH at the initial price of 2,000 USDC, the trader receives less because the trade itself changes the pool's reserves. This difference between the initial spot price and the realized execution price is known as slippage.

Table 1 shows a simple constant-product example: a trader adds 2,000 USDC to a pool holding 10 ETH and 20,000 USDC, receives 0.91 ETH, and pays an execution price above the initial price of 2,000 USDC per ETH.

Table 1 — Simple constant-product AMM example

Step	Calculation	Result
Initial ETH reserve	x	10 ETH
Initial USDC reserve	y	20,000 USDC
Constant product	$x \times y = k$	$10 \times 20{,}000 = 200{,}000$
Initial implied price	20,000 / 10	2,000 USDC/ETH
Trader adds USDC	20,000 + 2,000	22,000 USDC
New ETH reserve	200,000 / 22,000	9.09 ETH
ETH received by trader	10 − 9.09	0.91 ETH
Realized execution price	2,000 / 0.91	≈ 2,198 USDC/ETH

Source: prepared by the author.

Liquidity providers earn a share of the trading fees generated by swaps executed through the pool. This created a new source of on-chain yield that differs from traditional lending activity. Instead of lending assets directly to borrowers, liquidity providers earn income by making assets available for trading. During the period commonly known as “DeFi Summer,” liquidity provision was frequently combined with token incentives, giving rise to yield farming strategies in which users actively moved capital across protocols seeking higher returns.

However, liquidity provision also introduced specific risks. One of the most important is divergence loss, more commonly known in DeFi as impermanent loss. This occurs when the relative price of the assets in the pool changes after liquidity is provided. Because the AMM automatically rebalances the pool through trades and arbitrage, the liquidity provider ends up holding a different composition of assets than they would have held outside the pool. If the price movement is large, the LP position may be worth less than a passive holding strategy, even after accounting for trading fees [16].

Table 2 shows the same pool after the external price of ETH doubles to 4,000 USDC: arbitrage adjusts the reserves, and the liquidity provider ends up with 56,568 USDC in value against 60,000 USDC if the assets had simply been held, a loss of 5.72%.

Table 2 — Simple impermanent-loss example

Step	Calculation	Result
Initial ETH reserve	x	10 ETH
Initial USDC reserve	y	20,000 USDC
Initial ETH price	20,000 / 10	2,000 USDC/ETH
Constant product	x × y = k	200,000
New external ETH price	—	4,000 USDC/ETH
New pool condition	y / x = 4,000	pool adjusts by arbitrage
New ETH reserve	√(200,000 / 4,000)	7.07 ETH
New USDC reserve	4,000 × 7.07	28,284 USDC
LP pool value after price change	7.07 × 4,000 + 28,284	56,568 USDC
Value if LP had simply held assets	10 × 4,000 + 20,000	60,000 USDC

Step	Calculation	Result
Divergence loss / impermanent loss	56,568 / 60,000 − 1	−5.72%

Source: prepared by the author.

In this example, the LP does not lose money in absolute terms: the pool position increases from 40,000 USDC to approximately 56,568 USDC. However, the LP would have had 60,000 USDC by simply holding the original 10 ETH and 20,000 USDC outside the pool. The difference between these two outcomes is divergence loss. The loss is described as impermanent because it is not realized while the position remains in the pool: if the external price returns to the level at which liquidity was provided, the divergence disappears. It becomes permanent only when the liquidity provider withdraws at a price different from the initial one.

The previous example illustrated slippage from the trader's perspective: the trader received less ETH than expected because the trade moved the pool along the pricing curve. This example illustrates impermanent loss from the liquidity provider's perspective: after the external ETH price changed, arbitrage rebalanced the pool, leaving the LP with a different asset composition than a passive holding strategy.

Table 3 summarizes the difference between the two effects: slippage falls on the trader, as a result of the trade itself, while impermanent loss falls on the liquidity provider, as a result of a price movement outside the pool.

Table 3 — Slippage and impermanent loss: different perspectives

Concept	Perspective	What changes
Slippage	Trader	The trade changes the execution price
Impermanent loss / divergence loss	Liquidity provider	External price movement changes the pool composition

Source: prepared by the author.

The constant-product model is not the only design in use. Two later developments are relevant for the discussion that follows. Curve introduced a pricing function intended for assets that are expected to trade close to parity, such as two stablecoins or a staking derivative and the asset it represents. Its curve is deliberately flat around the equilibrium point, so that ordinary trades between closely related assets produce very little price impact, and only becomes steep when the pool moves far from that point.

Uniswap V3 introduced concentrated liquidity, allowing liquidity providers to allocate capital to a specific price range instead of the entire curve. Within that range the pool behaves as if it held far more capital, which reduces slippage and increases fee income per unit deposited. The trade-off is that the position stops earning fees once the price leaves the chosen range, and the exposure to divergence loss becomes more concentrated.

Both developments point in the same direction: the pricing function of an AMM can be designed around the economic relationship between the pooled assets, rather than treating every pair as a pair of unrelated assets. This principle is relevant to the analysis of Pendle in Chapter 4, where the pooled assets are linked by a maturity date rather than by a peg or a price range.

3.3 Staking and Restaking

The transition of major blockchain networks toward Proof-of-Stake (PoS) consensus mechanisms introduced staking as a major source of yield generation within DeFi.

In staking systems, users lock digital assets in order to participate in network validation and security processes. In return, validators and delegators receive staking rewards. Ethereum's transition to Proof-of-Stake significantly accelerated the importance of staking-based yield within the broader DeFi ecosystem.

These tokenized claims deliver their yield in different ways. Some, such as stETH, are rebasing tokens: the holder's balance grows as staking rewards accumulate, while each unit remains close to one ETH in value. Others keep the holder's balance fixed and accrue rewards through a rising exchange rate against the underlying, so that each token becomes redeemable for progressively more of it over time. This is the model of wstETH and of the interest-bearing tokens issued by lending protocols. The economic position is the same in both cases; what differs is the accounting interface through which the yield is delivered. This heterogeneity matters for what follows: it is precisely what Pendle's Standardized Yield layer, examined in Section 4.2, is designed to absorb before the principal and the yield can be separated [11].

The development of liquid staking protocols such as Lido introduced Liquid Staking Tokens (LSTs), allowing users to maintain liquidity while continuing to earn staking rewards. Instead of locking ETH directly, users receive tokenized claims such as stETH representing their staked positions.

More recently, restaking protocols such as EigenLayer expanded this structure further. Through restaking, previously staked assets may be reused to secure additional services and infrastructures, creating new forms of yield generation. This process gave rise to Liquid

Restaking Tokens (LRTs), which represent increasingly layered yield structures within DeFi [7].

As a result, DeFi yields became progressively more complex, incorporating multiple interconnected layers of protocol dependency, validator exposure, collateral structures, and incentive systems.

3.4 Composability and Layered Yields

One of the defining characteristics of DeFi is composability, often described as the ability of decentralized protocols to interact with one another as modular financial building blocks.

Unlike traditional financial systems, where products are frequently isolated within institutional infrastructures, DeFi protocols can be combined recursively through smart contracts. This enables the creation of layered financial structures in which yield generation depends simultaneously on multiple interconnected protocols.

For example, a yield-bearing position traded through Pendle may ultimately derive returns from several distinct layers, including:

1. Staking rewards from Ethereum validators: yield generated by validators that help secure Ethereum's Proof-of-Stake network. Users earn rewards for contributing ETH to network validation.

2. Liquid staking protocols such as Lido: protocols that allow users to stake ETH while receiving a liquid token, such as stETH, representing the staked position plus accrued staking rewards.

3. Restaking infrastructures such as EigenLayer: systems that allow already-staked ETH or liquid staking tokens to be reused to secure additional services, creating extra yield but also adding new risk layers.

4. Lending activity: yield generated when users supply assets to lending protocols such as Aave or Morpho and earn interest paid by borrowers.

5. Liquidity provision: yield generated by providing assets to decentralized exchange liquidity pools, earning trading fees from swaps executed through the pool.

6. Governance incentives: rewards or benefits linked to protocol governance, often used to encourage users to participate, vote, or allocate liquidity to specific pools.

7. Protocol emissions or points systems: additional rewards distributed by protocols to attract users and liquidity. These may be token emissions, incentive campaigns, or points that may later convert into rewards.

As a result, tokenized yield in DeFi often represents a composite stream of returns rather than a single isolated source of income [9].

This layered architecture creates important differences between DeFi yield markets and traditional fixed income systems. In traditional bonds, coupon payments generally derive from the issuer's contractual obligations and credit quality. In DeFi, however, yields may depend on dynamic interactions between smart contracts, collateral systems, liquidity conditions, validator performance, and protocol incentives. Therefore, understanding composability and layered yields is essential for interpreting the economic meaning of tokenized yield markets and the implied rates observed in protocols such as Pendle.

Yield tokenization may also be compared, with caution, to the logic of securitization in traditional finance. In securitization, future cash flows from underlying assets are pooled, structured, and transformed into tradable securities. Similarly, in DeFi yield tokenization, future yield streams generated by on-chain assets can be separated from the principal and traded independently. However, the analogy is limited: yield tokenization is not securitization in the traditional legal sense: there is no special purpose vehicle, no legal assignment of receivables, no credit tranches, no rating, and no traditional legal contract. Instead, the structure operates entirely through smart contracts. It shares the broader financial logic of transforming future cash flows into tradable instruments without replicating the institutional apparatus that surrounds securitization in traditional markets. [38].

At first sight, DeFi composability may appear unusually risky or excessive because it makes the layering of claims, collateral, and financial functions highly visible. However, the broader economic logic is not entirely foreign to traditional finance. There is a partial analogy between DeFi composability and the logic of money multiplication in traditional banking, since in both systems an initial economic base of assets or collateral may support multiple layers of claims, exposures, or financial uses. In banking, this process is institutionally organized through deposit creation, credit intermediation, reserve structures, and maturity transformation. In DeFi, by contrast, composability operates through the reuse of collateral, staking positions, and tokenized claims across programmable and open-source protocols. In both cases, greater capital efficiency may come together with greater fragility under stress. DeFi therefore does not invent

the logic of financial layering, but reconfigures it within open-source, programmable, and decentralized infrastructure [39].

An important distinction, however, lies in how losses are allocated under stress. In traditional financial systems, especially when large intermediaries operate under the expectation of institutional support or public rescue, the socialization of losses may generate moral hazard by weakening market discipline and encouraging excessive risk-taking [40]. In decentralized systems, by contrast, failures tend to be more directly internalized by the participants who assumed the underlying risks.

Part of the initial reaction to DeFi may therefore reflect the strangeness, the transparency, and the lack of familiarity of the ecosystem, rather than a difference of economic nature. The logic of layering, of reuse of the economic base, and of the transmission of fragility is not exclusive to DeFi; what changes is where these dynamics become visible and how they are ultimately resolved. The interpretive move made here is to read DeFi's absence of institutional backstops not as an unusual fragility but as a structural difference in loss allocation, whose evaluation depends on which trade-off one considers preferable.

4 YIELD TOKENIZATION AND THE CASE OF PENDLE

Chapter 3 described how yield is generated in DeFi, explained how a general-purpose Automated Market Maker works, and introduced the concepts of the protocols that preceded Pendle. This chapter examines Pendle, the case study of this dissertation, through its whitepapers and documentation, which show how the protocol's designers translated fixed income concepts into code. It explains each of the parts the protocol uses to assemble a complete market around those concepts, and compares these parts with the earlier protocols from which they were adapted, showing what had to change for fixed income concepts to be built in. Throughout, it identifies where the analogy with traditional fixed income holds and where it breaks down.

4.1 Pendle Overview

Pendle is a decentralized protocol focused on yield tokenization within DeFi. It allows the future yield generated by a yield-bearing asset to be separated from the underlying principal and traded as an independent claim. This construction distinguishes Pendle from previous DeFi yield mechanisms (lending protocols, staking platforms, liquidity provision, and automated vault strategies), which produce yield as an outcome of protocol activity but do not treat that yield as a tradable financial object in its own right.

The protocol emerged within a broader context in which DeFi participants increasingly sought more predictable forms of yield exposure and greater flexibility in managing future returns. While early DeFi protocols primarily generated floating and incentive-driven yields (variable, difficult to hedge, and dependent on real-time market conditions), Pendle introduced mechanisms that allow market participants to obtain fixed-yield exposure, hedge yield expectations, or trade future yield independently from ownership of the underlying asset. The problem the protocol addresses is therefore not the absence of yield in DeFi, but the absence of instruments through which yield itself can be priced, hedged, and traded as an independent object.

This innovation became particularly relevant with the growth of staking, liquid staking, restaking, and yield-bearing stablecoin ecosystems, where users increasingly hold assets that continuously generate on-chain yield streams. The volume of such assets has grown substantially, and their yield streams have become a category of value large enough to sustain

a dedicated market. Pendle is the leading protocol operating this market, and its architecture (the mechanics of Principal and Yield Tokens, the specialized Automated Market Maker, the layered composition of the yields it tokenizes, and the resulting comparison with traditional fixed income) is examined in the sections that follow.

4.2 Principal Token (PT), Yield Token (YT), and Standardized Yield (SY) Mechanics

The core mechanism underlying Pendle is the separation of a yield-bearing asset into distinct tradable components representing principal and future yield. This process, referred to as yield tokenization, allows market participants to isolate and trade different economic exposures associated with an underlying yield-generating position. To facilitate this structure, Pendle introduces three central token types: the Standardized Yield (SY), the Principal Token (PT), and the Yield Token (YT).

The Standardized Yield functions as the protocol's standardized representation of yield-bearing assets. Since DeFi yield sources may originate from multiple protocols and mechanisms (including staking, liquid staking, restaking, lending, or yield-bearing stablecoins), SY provides a unified framework that allows heterogeneous yield-generating assets to interact within Pendle's infrastructure [11]. Each SY unit is associated with an exchange rate against its underlying that grows continuously as yield accrues, providing a common measure of yield accumulation across otherwise incompatible sources.

Once deposited into the protocol, the SY position can be separated into two independent components: the Principal Token and the Yield Token.

The Principal Token represents a claim on the underlying asset at maturity. Holders of PTs are entitled to redeem the underlying asset when the maturity date is reached, regardless of the future yield generated during the holding period. Economically, PTs exhibit characteristics functionally analogous to zero-coupon bonds, since they can typically be purchased at a discount relative to their redemption value and converge to the underlying at expiration.

The Yield Token represents the right to receive the future yield generated by the underlying asset until maturity. Unlike PTs, YTs provide exposure specifically to the future income stream associated with the yield-bearing position. As a result, YTs allow market participants to speculate on, hedge, or trade future yield expectations independently from ownership of the principal itself.

This separation between principal and future yield represents one of the most important financial innovations introduced by Pendle. In traditional finance, future cash flows are often embedded within the same instrument. In Pendle, future yield becomes an independently tradable financial object. As a result, the protocol introduces mechanisms that resemble certain functions traditionally associated with fixed income markets, including discounted pricing, maturity structures, implied yield formation, and the separation between principal and future cash flows.

Yield tokenization in Pendle can be compared to the mechanism of coupon stripping applied to bonds in traditional fixed income markets, in which a coupon-bearing bond is decomposed into a zero-coupon bond representing the principal and separate instruments representing each future coupon payment. Both components are traded independently and derive their value from the same underlying obligation, but neither subordinates the other. Pendle's Principal Token performs the role of the stripped principal (analogous to a STRIPS Treasury), while the Yield Token performs the aggregate role of the stripped coupons, with the important qualification that the yield stream in a Pendle position is variable in rate and duration, whereas coupons in a traditional bond are typically fixed at issuance.

A further qualification concerns the cash flow structure of the Yield Token relative to stripped coupons in traditional fixed income. Stripped coupons pay at defined dates and in fixed amounts, since they represent specific coupon payments detached from a bond whose coupon schedule was fixed at issuance. The Yield Token, by contrast, distributes yield continuously as it accrues, and the total amount received by the holder depends on how much yield the underlying asset actually generates (realized yield) between the moment of purchase and the maturity date, not on any amount fixed in advance (implied APY). The two instruments perform the same economic role, isolating the future income component of a yield-generating position and making it independently tradable, but through structurally different cash flow generators. Recognizing this difference is important to avoid overextending the coupon-stripping analogy: the function is equivalent, the mechanism is not.

A more specific caveat applies to the PT-as-zero-coupon-bond analogy and deserves explicit treatment, since it is central to the analytical position of this dissertation. The analogy holds rigorously only when the underlying yield-bearing asset is denominated in a stable unit of account, typically a high-quality stablecoin such as USDC or USDT wrapped through the SY interface. In such cases, a PT purchased at a discount and held to maturity redeems for a defined amount of the underlying whose fiat value is predictable at the time of purchase, and

the position provides fixed-yield exposure both in the underlying unit and in fiat terms. This is the same structure that defines a traditional zero-coupon bond in a currency-stable environment.

When the underlying is a volatile asset, such as staked ETH represented by stETH or a liquid restaking token, the situation is different. A PT on such an underlying still converges to a defined number of underlying tokens at maturity, and it still provides fixed-yield exposure in terms of that underlying. But the fiat value of the redemption amount is not fixed, because the price of the underlying can move substantially between purchase and maturity. A PT-stETH position held to maturity delivers one stETH per unit as promised, but the dollar value of that stETH depends on the price of ETH at redemption. In this second case, the PT reproduces the *fixed-quantity* property of a zero-coupon bond without reproducing the *fixed-value* property that most fixed income analysis implicitly assumes. Interpreting a PT on a volatile underlying as a zero-coupon bond in the strict sense is therefore analytically misleading, and any comparison with traditional fixed income concepts should acknowledge this distinction explicitly.

It is important to notice that this structure is not unique to DeFi. A directly analogous instrument exists in traditional fixed income: sovereign or corporate bonds denominated in a foreign currency relative to the investor's unit of account. A U.S. dollar-denominated Treasury bond held by a European investor delivers a fixed cash flow in dollars but a variable cash flow in euros, whose realized value depends on the EUR/USD exchange rate at each coupon and at maturity. The instrument is fixed income in relation to its own unit of denomination and simultaneously carries foreign exchange exposure in relation to any other unit of account. A PT-stETH position operates on the same principle, with ETH taking the role of the denomination unit: the position delivers a fixed quantity of stETH at maturity but its value in any other unit (dollars, euros, or another crypto-asset) depends on the price of the underlying at redemption. Interpreting a PT on a volatile underlying therefore requires the same distinction that has always applied to fixed income instruments denominated in a currency different from the investor's reference unit: the fixed-quantity property is preserved by construction, while the fixed-value property depends on which unit of account the analysis is conducted in.

The process can be summarized conceptually as follows: a yield-bearing asset is wrapped into Standardized Yield form, and the resulting SY is then split into a Principal Token and a Yield Token. Formally:

$$Yield - Bearing\ Asset \rightarrow SY \rightarrow PT + YT$$

Where SY standardizes the yield-bearing position; PT represents the principal redeemable at maturity; and YT represents the future yield stream generated until maturity. This identity:

$$PT + YT = SY$$

is a mathematical consequence of the split mechanism and holds at any point in the life of a Pendle market. It is the on-chain equivalent of the accounting relationship between the stripped principal and stripped coupons of a decomposed bond in traditional fixed income infrastructure.

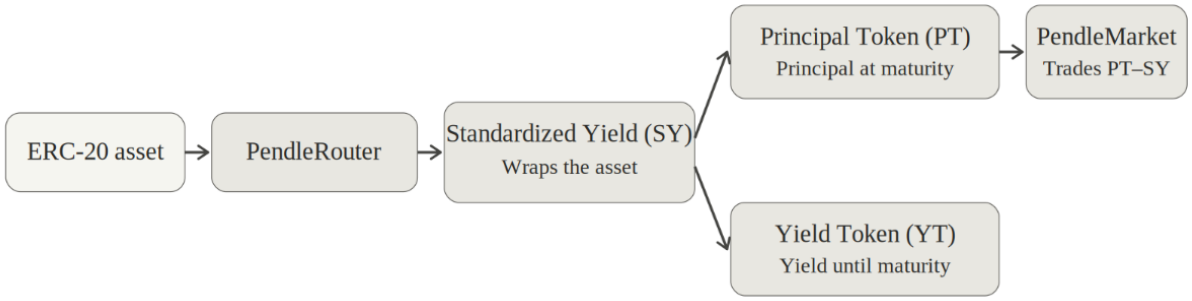


Figura 1 — Simplified Architecture of Pendle. The user deposits an ERC-20 yield-bearing token through the PendleRouter. The token is wrapped into an ERC-5115 Standardized Yield token (SY), which is then decomposed into a Principal Token (PT), redeemable for the underlying at maturity, and a Yield Token (YT), representing the future yield stream. The PT is traded against the SY within the PendleMarket. Source: prepared by the author, based on Pendle Finance Documentation.

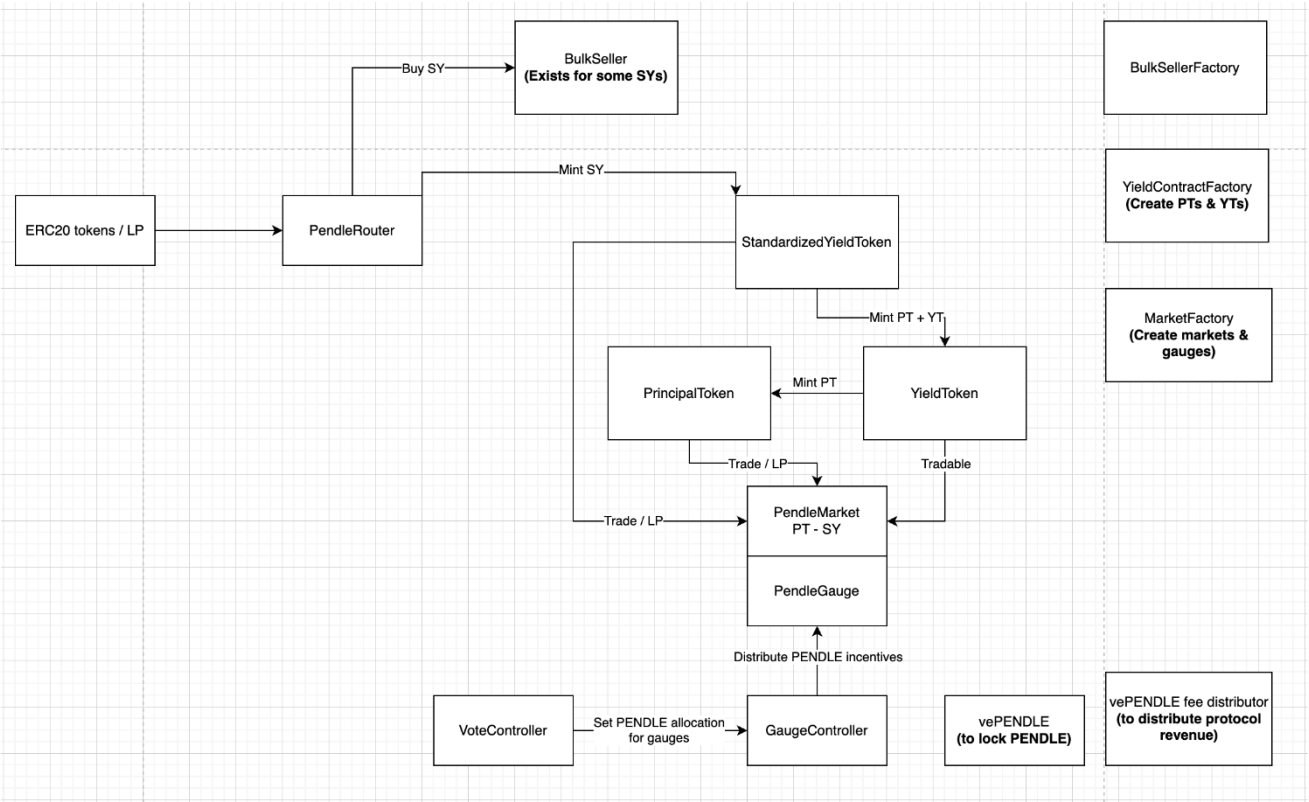


Figura 2 — High-Level Architecture of Pendle V2. Source: Pendle Finance Documentation. Available at: https://docs.pendle.finance/pendle-v2-dev/HighLevelArchitecture. Accessed: 29/07/2026.

The figure above shows the complete architecture of the Pendle V2 protocol, connecting all the main smart contracts that operate within the system.

A user starts by depositing an ERC-20 yield-bearing token (or a Liquidity Provider token) into the PendleRouter, which is the entry point of the protocol. The Router forwards the deposit to the corresponding StandardizedYield (SY) contract, which wraps the underlying asset into the standardized SY form. From the SY, the protocol mints two new tokens together: the Principal Token (PT) and the Yield Token (YT).

The PT and the SY are then traded within the PendleMarket, which is the market contract of the protocol. The YT is not traded directly against the SY, but it remains tradable through internal mechanisms of the market. When a user provides liquidity, the LP tokens received are also linked to the PendleGauge, which is the contract responsible for distributing PENDLE incentives to liquidity providers.

The figure also shows the governance layer of the protocol. Users who lock their PENDLE tokens receive vePENDLE, which gives them voting rights over how the protocol distributes incentives. Through the VoteController, vePENDLE holders decide the PENDLE allocation across the different gauges, and the GaugeController executes those decisions. In addition,

vePENDLE holders receive a share of the protocol revenue through the vePENDLE fee distributor.

Finally, three factory contracts are responsible for deploying new components of the protocol. The BulkSellerFactory deploys BulkSellers, which provide an alternative path for users to acquire SY tokens; the YieldContractFactory deploys new PT and YT contracts; and the MarketFactory deploys new Pendle markets and their associated gauges.

Together, these components describe the full operational architecture of Pendle V2: the flow of user deposits, the tokenization mechanism, the trading and liquidity layer, the governance system, and the factories that create the protocol's structure.

4.3 Implied Rates and Term Structure

One of the most significant innovations introduced by Pendle is the possibility of obtaining fixed-yield exposure through the pricing dynamics of Principal Tokens. By separating the principal component of a yield-bearing asset from its future yield stream, Pendle creates a market in which implied rates emerge directly from the relationship between PT prices, redemption value, and maturity.

The economic logic underlying Principal Token pricing resembles certain characteristics traditionally associated with discounted fixed income instruments. A Principal Token can typically be purchased below its redemption value prior to maturity and later redeemed for the underlying asset at expiration. As maturity approaches, the PT price tends to converge toward the redemption value of the underlying asset.

Conceptually, prior to maturity, the following relationship holds:

$$PT\ Price\ <\ Redemption\ Value$$

This section examines how implied rates emerge from the pricing of Principal Tokens and how the coexistence of markets with different expiries produces an on-chain term structure of interest rates. An implied rate in Pendle is the annualized rate of return embedded in the current PT price relative to its redemption value at maturity. It is not a rate set by any issuer or agreed on at the moment of purchase; it emerges from market equilibrium in the AMM at each point in time. As the PT price moves, the implied rate moves inversely, in the same way that yield and price move inversely for a zero-coupon bond in traditional fixed income.

The pricing of a Principal Token can be expressed through the same mathematical relationship that defines the present value of a zero-coupon bond in traditional fixed income. If the PT redeems for one unit of the underlying at maturity, its market price today reflects a discount factor determined by the implied rate and the time to maturity, following the relationship:

$$PT\ price\ =\ e^{-rt}$$

where r is the implied continuously compounded rate and t is the time remaining until expiry. This is the same discount formula that applies to a stripped Treasury or any zero-coupon claim in fixed income, and it reinforces that the PT is not merely analogous to a zero-coupon bond by convention: its pricing follows the same underlying mathematics.

The way this rate is obtained requires an important distinction between the two markets. In traditional fixed income, a bond usually pays coupons and has a face value defined at issuance. From its market price, you can compute a single rate called the yield to maturity (YTM), which is the rate that discounts all its future cash flows back to the current price. If you plot the YTMs of bonds with different maturities, you get a yield curve. However, this YTM curve mixes the effect of the coupons with the effect of the principal, so it is not the ideal reference for pricing or for describing the term structure. To get a cleaner curve, traditional fixed income uses a procedure called bootstrapping to extract a separate curve composed only of zero-coupon rates, one for each maturity. This is called the spot rate curve, and it is the reference used to describe the term structure of interest rates.

In Pendle this extra step does not exist. Because the protocol already splits the yield-bearing asset into PT and YT on-chain, the PT is a zero-coupon claim by construction. The implied rate extracted from the PT price is therefore already a spot rate, and the curve formed by markets with different expiries is already a spot rate curve. So the concept of a term structure is the same in both markets, but the way it becomes visible is different: in traditional fixed income you have to reconstruct it through bootstrapping, while in Pendle you can read it directly from the PT prices, one point per market.

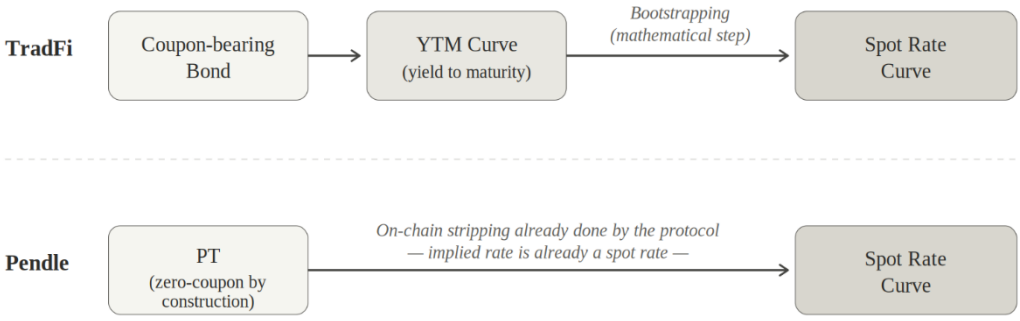


Figura 3 — How the Spot Rate Curve is Obtained: TradFi and Pendle. In traditional fixed income, coupon-bearing bonds yield a Yield-to-Maturity (YTM) curve directly readable from bond prices; extracting the pure Spot Rate Curve requires an intermediate mathematical step known as bootstrapping. In Pendle, this stripping step is performed on-chain by the protocol at the moment a yield-bearing asset is decomposed into Principal Token and Yield Token. Because each PT is by construction a zero-coupon claim, its implied rate is already a spot rate, and the resulting curve is the Spot Rate Curve directly, without an intermediate reconstruction. Source: prepared by the author.

A related property that follows from this pricing relationship is that a Principal Token has duration equal to its time to maturity, in the same way that a zero-coupon bond does. Duration, in fixed income, measures how sensitive the price of an instrument is to a change in the interest rate used to discount its future cash flows. For a zero-coupon instrument, one that pays no intermediate cash flows and returns its full value only at expiry, this sensitivity equals exactly the time remaining until maturity. A Principal Token, being a zero-coupon claim by construction, inherits this property directly: its price responds to changes in the implied rate according to the same rules that govern any zero-coupon fixed income instrument. Recent academic work has begun to formalize the duration and convexity properties of tokenized fixed income positions such as Pendle's Principal Token, extending traditional fixed income risk measures to the on-chain context [8.2].

A further structural difference concerns the treatment of maturity. In traditional fixed income, the maturity of a bond is decided by the issuer at the moment of issuance. The term structure of interest rates observed in the market (the sovereign yield curve, for instance) emerges from the coexistence of bonds issued by the same or comparable issuers at different maturities. The U.S. Treasury issues securities across a range of maturities, from short-term bills to long-term bonds, and it is the yields observed on these different issues, taken together, that compose the sovereign yield curve.

In Pendle, maturity is not an issuer decision but a parameter of the smart contract that defines each individual market. Each market is created by the Pendle governance through the MarketFactory contract, which deploys a new market for a specific pairing of underlying and expiry date (Figura 4). Once deployed, the market operates independently: it holds its own liquidity pool of PT and SY, mints its own PT and YT tokens, and produces its own implied rate emerging from the price at which those tokens trade against each other. On a single underlying such as stETH, several markets may therefore exist simultaneously, for example PT-stETH-DEC2025, PT-stETH-MAR2026, and PT-stETH-JUN2026. When a market reaches its expiry date, it is closed and PT holders redeem for the underlying; new markets are typically deployed for future dates to preserve continuity in the term structure.

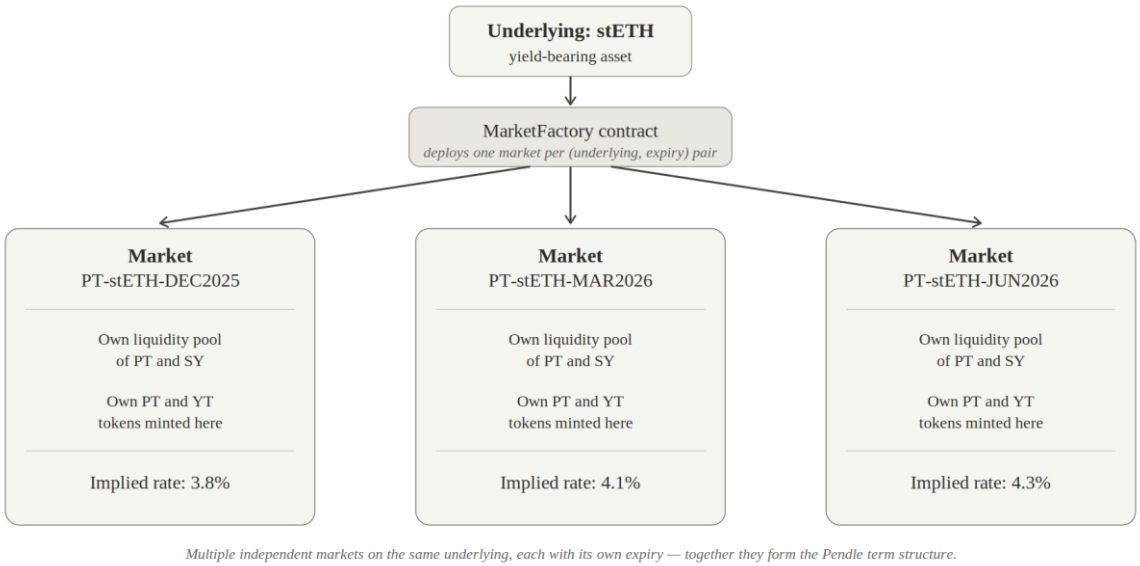


Figura 4 — Multiple Pendle Markets on the Same Underlying. A single yield-bearing asset (in this example, stETH) can be tokenized by multiple Pendle markets simultaneously. Each market is deployed by the MarketFactory contract for a specific expiry date and operates independently, holding its own liquidity pool of PT and SY, minting its own PT and YT tokens, and producing its own implied rate. The coexistence of these markets across different expiries forms the Pendle term structure of interest rates for the underlying. Source: prepared by the author, based on Pendle Finance Documentation.

A term structure emerges in Pendle from the coexistence of these markets on the same underlying, rather than from a single issuer distributing debt across a maturity spectrum. The observed implied rates across markets with different expiries form an empirical yield curve for that underlying, a point-by-point curve rather than a parametric continuous one. When a market reaches its expiry, it is closed, and new markets are typically created for future dates to preserve the term structure.

The implied rates observed across Pendle markets are interest rates in the same mathematical sense as those observed as spot rates in traditional fixed income: they function as discount factors that bring future cash flows to present value, and they form yield curves that reflect market expectations for each maturity. The two curves differ in what they capture economically. A traditional sovereign yield curve is anchored in a benchmark risk-free rate, typically the shortest-term government security, against which all other rates are read as spreads. In DeFi, no equivalent risk-free anchor exists: there is no sovereign issuer, no monetary authority, and no institutionally guaranteed default-free instrument. Each Pendle curve therefore floats without an exogenous reference and reflects only the expected yield of its own underlying. Ongoing research has begun to address this gap: some works propose inferring implicit interest rate curves from cryptocurrency derivatives markets [10], while more recent proposals aim to build fixed-income AMM designs capable of supporting explicit term structure construction on-chain [30][31].

A related distinction concerns the interpretive weight that can be placed on these implied rates. Sovereign yield curves serve as universally accepted benchmark references used for valuation and risk management across the entire fixed income system. Pendle's implied rates do not play this role: they are protocol-specific, reflecting the yield-generating mechanisms and risks of the underlying DeFi ecosystem, including staking rewards, restaking exposure, liquidity conditions, protocol incentives, smart contract risk, depeg risk, and governance structures. Pendle therefore does not create a universal crypto-native risk-free curve; it creates markets for protocol-specific implied rates derived from tokenized yield positions.

Beyond the absence of a risk-free anchor, the two curves also differ in origin and structure. Traditional sovereign curves are composed of instruments issued by a single institutional entity that distributes debt across maturities, producing a continuous curve typically fitted through parametric models such as Nelson-Siegel. The Pendle curve is composed of independent markets deployed by protocol governance, each with its own smart contract and its own maturity, producing an empirical point-by-point curve whose density depends on how many active markets exist for a given underlying. Sovereign curves also extend to horizons of thirty years or more, while Pendle curves typically span months to a year or two.

Despite these structural differences, the analytical tools developed for traditional yield curves apply directly to Pendle. Forward rates, the implied rates for future periods between two known maturities, can be extracted between any two markets on the same underlying using the same mathematics used in traditional fixed income; inconsistencies between these forward rates and the observed yield dynamics of the underlying create arbitrage opportunities. The slope and

shape of the curve convey information about market expectations regarding the evolution of the underlying yield, analogous to how the slope of a sovereign curve conveys expectations of monetary policy. Parametric curve-fitting models originally developed for sovereign markets can also be adjusted to Pendle curves, subject to the caveat that the number of observation points is comparatively small.

Quadro 2 — Analytical Tools Applied to Yield Curves: TradFi and Pendle

Analytical tool	**What it does**	**In sovereign fixed income**	**In Pendle**
Forward rate extraction	Derives the implied rate for a future period between two known maturities on the same curve.	Extracted from Treasury yields at different maturities to infer expected short rates and to price forward-starting instruments.	Extracted between two Pendle markets on the same underlying (e.g., between PT-stETH-DEC2025 and PT-stETH-MAR2026) to infer the market's expected yield for the intervening period.
Slope analysis	Reads the difference between short-term and long-term rates on the curve as a signal of market expectations.	A positively sloped Treasury curve signals expectations of rising rates; a flat or inverted curve signals expectations of falling rates or economic contraction.	A positively sloped Pendle curve on a given underlying signals expectations of rising yield for that underlying; a flat or inverted curve signals expectations of stable or falling yield.
Shape analysis	Reads the curvature of the yield curve as a signal of expectations about specific future periods.	Humps or bends in the Treasury curve reflect expectations of monetary policy shifts at specific horizons.	Humps or bends in the Pendle curve reflect expectations of protocol-specific events at specific horizons (for example, an anticipated change in restaking yield).
Parametric curve fitting	Uses a mathematical model (such as Nelson-Siegel) to fit a continuous curve to a discrete set of observed rates.	Central banks and market participants routinely fit Nelson-Siegel or Svensson models to publish smooth Treasury curves from a limited set of observed securities.	The same models can be applied to Pendle curves, subject to the caveat that the number of active markets per underlying is small, limiting the number of observation points available for fitting.
Intertemporal arbitrage	Identifies inconsistencies between forward rates	Bond dealers exploit mispricings between implied	Arbitrageurs exploit mispricings between the

	implied by the curve and the observed dynamics of the underlying.	forward rates and expected realized rates through spread and butterfly trades.	forward rates implied by two Pendle markets and the observed yield generation of the underlying protocol.

Source: prepared by the author.

Additionally, the yield embedded within YTs may derive from multiple interconnected layers, including staking rewards, restaking systems, lending activity, liquidity provision, governance incentives, and protocol emissions. Consequently, the implied rates observed in Pendle markets reflect not only time value and market expectations, but also the layered and composable risk structure characteristic of DeFi. This point was developed in Section 3.4 and returns in Chapter 5.

4.3.1 The DeFi Yield Curve and the Traditional Yield Curve: Is Arbitrage Between Them Viable?

The previous sections showed that, for each underlying asset traded in Pendle, the market produces an implied rate for every maturity available. Taken together, the implied rates of one asset across its maturities form a yield curve: a line that shows how the rate changes with the time to maturity. Traditional finance also has yield curves, one built from government bonds and one for each class of credit risk, as Section 2.2 explained. This raises a natural question. If a position in DeFi carried the same risk as a corporate bond of a given rating, should its implied rate not equal the rate on that bond's yield curve? And if the two rates differed, would investors not lend where the rate is higher and borrow where it is lower, until the difference disappeared?

To answer this question, it helps to make explicit the reasoning behind it. In finance, two investments that carry the same risk and pay the same cash flows must have the same price, and therefore the same rate. If they did not, an investor could borrow through the cheaper one and lend through the more expensive one, and collect the difference. Because the two positions carry the same risk, the investor's exposure in one cancel the exposure in the other, and the difference is earned without taking on risk. This operation is called arbitrage. As more investors perform it, demand rises for the investment that pays more and falls for the one that pays less, and the two rates move toward each other until they meet. The principle that the rates must meet is known as the law of one price. Arbitrage is also what makes a market efficient, in the sense that it pushes prices to reflect all the information that investors have.

For arbitrage to bring the DeFi yield curve and a traditional yield curve together, three conditions must hold at the same time. The first is that the two positions carry the same risk, since this is what allows one exposure to cancel the other. The second is that capital can move freely from one market to the other, since arbitrage is performed by moving capital. The third is that the two investments pay the same thing, in the same currency, and that the payment comes from the same kind of source, that is, from the use of capital and not from a subsidy, since otherwise the investor is not comparing like with like. This section examines each condition in turn. It anticipates its conclusion here: none of the three conditions holds today, and the first fails for a reason that goes beyond the current stage of the market. The risks on the two sides are not of the same nature: on the traditional side, the risk is that a counterparty fails to keep a promise, while on the DeFi side, the risk is that code or a protocol fails to operate as intended. There is no common scale on which to measure the two. The two yield curves therefore need not converge. What arbitrage can produce between them is not equality but a price for their difference, that is, a risk premium.

The first condition: the same risk

The first condition for arbitrage is that a position in DeFi and a bond of a given rating carry the same risk. To examine whether this can be the case, it helps to recall that a bond carries four distinct risks. Credit risk is the risk that the issuer fails to pay. Interest rate risk arises when market rates rise after the bond is bought: new bonds pay more, the old bond becomes less attractive, and its price falls for anyone who needs to sell before maturity. Liquidity risk is the risk of finding few buyers at the moment of selling, so that the seller has to accept a lower price. Currency risk is the risk that the bond pays in a currency that loses value. A credit rating measures only credit risk. When one speaks of a bond “of rating A”, one is speaking of credit risk alone.

Three of these four risks have a direct counterpart in a Principal Token. A PT is bought at a discount to its redemption value, and the implied rate is the size of that discount expressed as a yearly rate. If the implied rate rises, the discount becomes larger, and the price of the PT falls. This is interest rate risk in the same form as in a bond. A PT can also be hard to sell without moving its price when the pool in which it trades holds little liquidity, which is liquidity risk. And it pays in the currency of its underlying asset, which may be ETH or a tokenized dollar, which is currency risk. In these three dimensions the analogy with fixed income holds, and arbitrage between the two markets is possible in principle.

It is in credit risk that the analogy stops. A PT has no issuer, so there is no credit risk in the traditional sense. Redemption at maturity is not a promise kept by a counterparty but an

operation executed by code. In its place, the PT carries risks of a different nature: the contract may contain a flaw, the protocol that generates the yield may stop operating, and the underlying asset may lose value. The protection against these risks also comes from a different source.

Table 5 summarizes the correspondence between the four risks of a bond and their counterparts in a Principal Token.

Table 5 — Risks of a bond and their counterparts in a Principal Token

Risk in a bond	Counterpart in a Principal Token	Does the analogy hold?
Credit risk: the issuer fails to pay	None. Replaced by contract and protocol risk, of a different nature	No
Interest rate risk: price falls when rates rise	Price falls when the implied rate rises	Yes
Liquidity risk: cost of selling without moving the price	Depends on the depth of the pool and of the order book	Yes
Currency risk: payment in a currency that loses value	Pays in ETH or in a tokenized dollar	Yes

Source: prepared by the author.

A bond is protected by the force of law and by institutions. A PT is protected by code that anyone can audit, by rules that are visible to all participants, and by execution that follows those rules automatically. Each system offers the protection that fits its own environment. A PT is not protected by institutions, in the same way that a bond is not protected by open code that anyone can verify. These are two systems of protection, not one system and its absence.

This is why the expression “the same risk” has no defined meaning when applied to a PT and a bond. A rating translates the financial health of an issuer into a probability of default, on a scale that the whole market accepts. No equivalent scale exists for translating a protocol’s exposure into a probability of loss. There are indicators, such as audits, time in operation, value deposited and history of incidents, but none of them is a common measure. An arbitrageur who could move capital freely between the two markets would still be unable to state that the risk is equal, because the two risks are not measured in the same language.

Markets compare risks of different natures all the time, and they do so without declaring them equal. They price the difference. The spread between a corporate yield curve and the government yield curve is such a price: the spread is the difference in rate between the two, measured in basis points, where one basis point is one hundredth of a percentage point. No one

claims that an AA issuer carries the risk of the Treasury, only that the difference is worth a certain number of basis points. The relation between a DeFi yield curve and a traditional yield curve, if it comes to exist in a stable form, will be of this kind. The two curves need not converge, because they price risks of different natures. What the maturation of the market may produce is a premium between them, and not their fusion.

The second condition: capital that moves freely

Arbitrage is performed by moving capital. To bring two rates together, an investor must be able to take funds from one market and place them in the other, in volume and repeatedly, until the difference disappears. If capital cannot cross from one market to the other, the two rates remain apart no matter how different they are.

Today, capital does not cross easily in either direction. Consider an institution that holds bonds, such as a pension fund or an insurance company, and that wishes to hold a Principal Token instead. Four obstacles stand in its way. Its mandate lists the assets it is allowed to hold, and a PT is usually not among them. It is required to keep its assets with a qualified custodian, while in DeFi whoever holds the private key holds the asset, which does not fit that requirement. It must identify the counterparty of each transaction, while in a liquidity pool there is no identifiable counterparty. And, in the case of a bank, an exposure to crypto-assets requires setting aside far more regulatory capital than an exposure to a bond, which makes the operation too costly to be worthwhile.

The opposite path is also closed. An investor holding capital in a stablecoin or in ETH cannot buy a bond directly. The funds must first be converted into fiat currency through a bank, an account must be opened with a broker, and identity and jurisdiction requirements must be met. Each step costs time and money and excludes part of the participants.

When capital cannot cross, each market forms its rate with the capital it holds within itself. The implied rate in DeFi reflects the supply and demand of those who are already in DeFi, and the bond rate reflects the supply and demand of those who are already in the bond market. The few participants able to operate in both, such as specialized funds, do not command the volume needed to align the curves. This situation is known as market segmentation: two markets that price separately because capital does not circulate between them.

Regulation is what determines whether capital crosses, and it can work in either direction. It can open the way, by establishing rules for custody and licensing that allow an institution to operate in DeFi within its obligations. It can also close it, by prohibiting the activity or making it too costly, including as a means of protecting the banking system from competition. The European MiCA regulation and the Dubai VARA framework, discussed in

Chapter 5, are two different answers to this same decision. Whether the second condition comes to hold is therefore not a question of market maturity alone, but of regulatory choice.

Even within DeFi, capital does not circulate perfectly. The same asset can carry different implied rates on different blockchains, and the differences remain open for longer than arbitrage would allow in a mature market. If arbitrage is incomplete within DeFi itself, it is more limited still between DeFi and the bond market.

The third condition: the same payment from the same kind of source

Arbitrage only makes sense if the two investments pay the same thing. When one investment pays in dollars and the other pays in ETH, the investor is not comparing two rates but two rates plus a bet on the exchange rate between the two currencies. To remove the bet, the investor would need to buy protection against currency movements, and the cost of that protection absorbs the difference between the rates. Traditional finance has a name for this: interest rate parity, the principle that rates in different currencies differ by what the market expects of the exchange rate between them. A PT on staked ETH pays in ETH, and its implied rate cannot be set against the rate of a dollar bond without passing through this correction.

The condition comes closest to holding when the underlying asset of the PT is a stablecoin. The payment is then in a tokenized dollar, and currency risk is reduced to the risk that the stablecoin loses its peg. This is the case in which a comparison with a dollar bond is most direct.

Even when the currency is the same, a second question remains: whether the payment comes from the same kind of source. The yield on a bond is payment for the use of capital. The borrower pays for the money it used. Part of the yield in DeFi has the same origin, such as the interest paid on a collateralized loan. Another part does not. Protocols distribute their own tokens to attract deposits, and these distributions are not payment for the use of capital but an incentive, paid in a currency that the protocol itself issues, with a volatile price and an uncertain duration. In Pendle this is visible in the pricing of Yield Tokens, which often reflects the expectation of receiving points or airdrops rather than the yield of the underlying asset alone. Comparing an advertised annual rate in DeFi with the rate on a bond mixes payment with subsidy.

A further difference lies in what forms the rate. The rate on a bond is formed by the supply of and the demand for credit, that is, by those who wish to borrow and those who wish to lend. The rate on a staking position is formed by the rules of the network: how many new tokens it issues and how many participants share them. These are different economic

mechanisms producing a number that goes by the same name. Two such rates may coincide, but they respond to different forces and move for different reasons.

The third condition therefore fails at three levels: the currency, the source of the payment and the mechanism that forms the rate. The first can be addressed with stablecoins. The other two cannot, because they are part of what yield in DeFi is.

4.4 Yield Tokenization and the case of Pendle

As described in Section 4.2, Pendle separates a yield-bearing asset into two distinct tradable claims. This section concerns the second of those claims: the Yield Token concentrates the variable component of the position: the right to receive the future yield generated by the underlying asset until maturity.

The yield accrues continuously and can be claimed by the holder at any moment before maturity. As a consequence, the value of the YT declines as maturity approaches. These two movements are part of the same process: on each day the holder receives the yield produced on that day, and the token comes to represent one day less of future yield. The value is not lost; it is transferred from the token to the holder. What remains inside the token at any point is only the yield that has not yet been paid.

A simple illustration clarifies the mechanism. Consider a YT purchased for 100 units with 180 days remaining until maturity. If the implied yield remains constant, after 90 days the token is worth approximately half of its initial price, because only half of the accrual period is left. At maturity the token is worth nothing, since no future yield remains to be claimed. If the yield collected over the whole period amounted to 120 units, the position produced a gain of 20 units.

This separation is economically significant because it transforms future yield from an embedded and non-isolated component of a yield-bearing asset into an independently tradable exposure. In a conventional DeFi position, the investor who holds a yield-bearing asset remains exposed simultaneously to principal risk and future yield variation. In Pendle, these exposures are split into two tokens, PT and YT. The PT holder gives up the future yield stream in exchange for a more predictable fixed-yield-like return, while the YT holder assumes the uncertainty associated with the future realized yield. The YT therefore represents the side of the market that absorbs the variability that the PT holder no longer wants to hold.

In this sense, the Yield Token performs a function comparable to interest rate derivatives in traditional finance. Traditional interest rate instruments such as forward rate agreements,

interest rate swaps, caps, floors, and swaptions allow market participants to isolate, transfer, and trade exposure to future interest rate movements. Pendle performs a similar economic operation within a crypto-native architecture. Instead of structuring exposure through bilateral contracts, dealer intermediation, benchmark-linked cash flows, or clearinghouse infrastructure, Pendle embeds the separation of principal and future yield directly into smart contracts and represents each side of the position as transferable on-chain tokens. One fundamental difference between the Pendle protocol and traditional finance is structural: the underlying asset from which the two tokens are minted remains locked in the contract until maturity and backs both positions simultaneously. PT and YT are therefore not two independent promises of payment, but two claims on the same collateral. In an interest rate swap, each payment depends on one counterparty transferring funds to the other, and fails if that counterparty defaults. In Pendle, both the yield and the principal come from the asset already deposited, so settlement does not depend on the solvency of a counterparty.

The economic logic can be understood through the relationship between expected yield and realized yield. The price paid for a YT reflects the market's expectation of the future yield to be generated by the underlying asset over the remaining period until maturity. If the realized yield over that period exceeds the yield implied by the YT purchase price, the YT position benefits. If the realized yield is lower than expected, the position underperforms. The magnitude of these outcomes follows from the price of the YT itself. Because the YT costs only a fraction of the value of the asset from which it was minted, the yield exposure obtained is considerably larger than the amount paid. Small differences between expected and realized yield therefore translate into large percentage gains or losses on the position. The YT holder is therefore exposed to future yield variation in a way that is economically analogous to the holder of an interest-rate-sensitive derivative position. The purpose is not merely to receive yield passively, but to take a view on the future path of yield generated by a specific on-chain asset or strategy.

A simple numerical illustration makes this relationship concrete. The example assumes that the identity PT + YT = SY holds at market prices, that the position is held until maturity, and abstracts from transaction fees. Consider a stablecoin-based market in which the underlying asset is worth 100 units, maturity is one year away, and the implied APY is 5%. The PT is priced at the discounted value of the principal, 100 divided by 1.05, or approximately 95.24 units. Since the two tokens together must equal the value of the underlying asset, the YT is priced at the remainder, approximately 4.76 units. That amount entitles the holder to the entire yield produced by 100 units of the asset over the year.

The outcome of the position depends on the yield the asset actually generates. If the realized yield is 5%, the holder collects 5 units against a cost of 4.76, and the position returns approximately the implied APY, the same return the PT holder secured at the outset. If the realized yield is 8%, the holder collects 8 units and gains 3.24 units, a return of roughly 68% on the amount invested. If the realized yield is 3%, the holder collects only 3 units and loses 1.76 units, a return of approximately −37%.

The asymmetry between these outcomes follows from the relationship between the price of the YT and the amount on which the yield accrues. The 4.76 units paid give exposure to the yield generated by 100 units, approximately twenty-one times the amount invested. A difference of three percentage points in realized yield therefore produces a change of several dozen percentage points in the return of the position. The exposure is nevertheless bounded: in the extreme case in which the asset generates no yield at all, the holder loses the 4.76 units paid and nothing more.

A stablecoin-based market was chosen for this illustration because it isolates the yield component. When the underlying asset is not a stablecoin, the price of the asset itself also varies, and the outcome of the position combines yield variation with price variation.

Tabela 3 — YT position outcomes under different levels of realized yield

Realized yield (%)	Realized yield (units)	Cost of YT	Net result	Return on outlay
0%	0.00	4.76	−4.76	−100%
3%	3.00	4.76	−1.76	−37%
5%	5.00	4.76	+0.24	+5%
8%	8.00	4.76	+3.24	+68%

Note: underlying asset of 100 units, one year to maturity, implied APY of 5%. Source: elaborated by the author.

Pendle reorganizes this logic in a different market environment. The YT does not reference a sovereign yield curve or a standardized benchmark rate. It references the realized yield of a specific DeFi yield-bearing asset. This yield may come from staking rewards, lending rates, liquidity provision, liquid restaking, stablecoin mechanisms, governance incentives, protocol emissions, points programs, or combinations of these sources. **As a result, the YT is not exposure to an economy-wide interest rate, but to a protocol-specific yield stream**. This distinction is central: Pendle creates tradable yield exposure, but the underlying yield is heterogeneous, composable, and dependent on the architecture of the protocols that generate it.

The PT/YT split therefore creates a two-sided market for yield expectations, and the YT is the instrument through which market participants can express a view on whether future realized yield will be higher or lower than what is currently priced by the market. If demand for YT increases, this may indicate that market participants expect future yield to be attractive relative to its current price. If demand for PT increases, this may indicate preference for fixed-yield-like exposure, lower uncertainty, or more conservative positioning.

These two sides also correspond to different market motivations. PT demand may be interpreted as a form of hedging demand, in the sense that participants seek to reduce uncertainty about future yield or lock in a more predictable return. YT demand, by contrast, may be interpreted as directional demand, in the sense that participants deliberately take exposure to the future path of yield. Hedging and directional motives therefore coexist within the same market: one side seeks protection from uncertain future yield, while the other side seeks exposure to it. This coexistence is similar to the logic of traditional interest rate derivative markets, where one participant often seeks protection against future rate movements while another is willing to take the opposite exposure.

The YT also concentrates a different risk profile from the PT. Because the YT holder does not hold the principal claim, the value of the position depends primarily on the yield generated before maturity. This makes YT exposure more sensitive to changes in yield expectations, protocol incentives, reward emissions, lending demand, staking conditions, restaking rewards, points campaigns, and market liquidity. If the yield stream declines, disappears, or becomes less valuable than expected, the YT may lose value rapidly. This loss, however, is bounded. The YT position is paid in full at the moment of purchase, so there is no margin requirement, no margin call and no liquidation. The least favourable outcome for the buyer is that the yield collected until maturity is worth less than the price originally paid. As maturity approaches, the remaining period over which yield can be earned also decreases, creating a **natural value-decay effect.** The YT is therefore structurally more exposed to uncertainty than the PT, because its value is tied to the future income component rather than the principal redemption component.

For all the reasons discussed above, the Yield Token is central to the interpretation of Pendle as a crypto-native fixed-income-like infrastructure. PTs alone would create a discounted principal instrument, but without YTs the future yield component would remain embedded in the underlying yield-bearing asset rather than becoming an independent market exposure. The PT/YT structure is therefore what allows Pendle to move beyond passive yield generation and

toward a more complete market architecture for pricing, transferring, and managing future yield risk on-chain.

4.5 Liquidity Engine: Pendle AMM Design and Order Book

4.5.1 An AMM Designed for Yield, Not for Spot Price

Pendle's AMM is not merely a liquidity mechanism; it is a market structure designed for pricing future yield exposure. This is the most important characteristic to establish before examining its construction. Traditional AMMs such as Uniswap or Curve are designed for spot asset trading and liquidity provision, and their pricing functions treat the assets in the pool as substitutable claims on present value. Pendle's AMM is designed for assets whose value is a function of time to maturity and expected future yield. As a result, its pricing dynamics depend not only on spot market conditions but on remaining time until maturity, expected future yield generation, discount convergence toward redemption value, protocol-specific risks, and market expectations regarding future on-chain returns.

This creates a significantly more sophisticated market environment than earlier DeFi infrastructures focused on liquidity mining or floating-yield farming. Pendle does not trade spot assets. It trades time, maturity, decay, and future yield expectations, and its AMM is built to express those variables as the central pricing signal.

A structural feature that follows from this design deserves specific mention. In a general-purpose AMM, liquidity providers face impermanent loss whenever the external price of the paired assets diverges from the entry ratio. In Pendle's AMM, this problem is eliminated by construction at maturity: because a Principal Token converges to one unit of the underlying at expiration regardless of the price path taken to get there, and because the pool pairs PT with its own SY, an LP position held to maturity ends with all of the LP's exposure denominated in the underlying. Impermanent loss does not accumulate as it would in a spot AMM, because the trajectory of PT price is bounded by its convergence property. This is one of the earliest indications that Pendle's AMM is a different kind of construct than Uniswap or Curve, one that draws its properties from the fixed income character of the instruments it trades rather than from the general-purpose logic of automated market making.

4.5.2 Design Lineage: From the Geometric Mean to Notional Finance

The Pendle V2 AMM design does not begin from scratch. The protocol adopts as its base the model developed by Notional Finance, an earlier protocol focused on fixed-rate on-chain lending through instruments called fCash, structurally analogous to zero-coupon bonds. Pendle adapts and generalizes this model beyond lending, applying it to any yield-generating asset in the ecosystem [17] [23] [28] [29].

The starting point is a simple but fundamental conceptual inversion. In any generic AMM, the price is the direct ratio between reserves, that is, how much of one asset exists relative to the other. For Principal Tokens, Pendle inverts this logic and defines the price of the asset in terms of PT, rather than the price of PT in terms of the asset.

The reason is financial. A PT has a rigid property: it converges to one unit of the underlying at maturity, regardless of any other consideration. This means the price of PT is always bounded between zero and one, and any movement in that price carries two mixed signals simultaneously: what the market expects about future yield, and simply how much time has passed. Working with the price of the underlying in PT terms, and extracting the annualized implied rate from it, separates the two signals cleanly. The same PT price implies completely different yields depending on how many days remain until maturity, and any AMM that ignores this temporal dimension prices the instrument incorrectly.

Two AMM designs preceded Pendle V2 and recognized the problem without fully resolving it. The geometric mean (used in Pendle V1 itself, and equivalent to Uniswap V2 with symmetric weights) produces a static curve in which time does not appear in the formula. A PT with one week to maturity and a PT with two years are priced by the same curve. As maturity approaches and arbitrageurs push the PT price toward one, the pool converges mechanically toward equal-weight inventory, but liquidity distributed elsewhere along the curve remains economically inactive: capital inefficiently allocated where no trade will happen.[17] [23]

The YieldSpace design improved on this by introducing time as an exponent in the pricing function, which produces a curve that converge correctly as time to maturity approaches zero. But the YieldSpace model has no free parameters: the same curve is applied to any asset, whether it is a stablecoin yielding four percent per year or a higher-risk asset yielding two hundred percent. There is no way to specialize the pool to the yield profile of the underlying asset, which leaves the liquidity inefficiently allocated for assets whose yields sit far from the curve's implicit centering.[28]

4.5.3 Operating in Rate Space: The Logit Function, rateScalar, and rateAnchor

The solution Pendle V2 adopts, based on the Notional Finance model, abandons the idea of starting from an invariant over reserves and defines the pricing formula directly as a function of proportion and time:

$$exchangeRate(t) = \frac{\ln\left(\frac{p(t)}{1-p(t)}\right)}{rateScalar(t)} + rateAnchor(t)$$

The term ln(p/(1-p)) is the logit function, the inverse of the sigmoid, which produces a curve that is flat in the center and explosive at the extremes. Financially, this means that ordinary trades near equilibrium move the implied rate very little (low slippage, high effective liquidity depth), while attempts to push the rate toward extremes require disproportionately large trades. This is precisely the behavior expected of a well-functioning fixed income market: rates should be sticky near their equilibrium level and resistant to being pushed to unrealistic levels by small trades.

On top of this base geometry, two parameters inject asset-specific financial intelligence into each pool. The rateScalar, defined as:

$$rateScalar(t) = \frac{scalarRoot}{yearsToExpiry(t)}$$

compresses the curve automatically as maturity approaches, concentrating liquidity where the market will actually operate. This is the analogue of duration: it governs the sensitivity of price to changes in the implied rate, and it decays as remaining time to maturity shrinks. The rateAnchor:

$$rateAnchor(t) = price(t) - \frac{\ln\left(\frac{p}{1-p}\right)}{rateScalar(t)}$$

Where:

$$price(t) = (1 + lastImpliedRate)^{yearsToExpiry(t)}$$

shifts the curve vertically to center the liquidity around the market's current equilibrium rate for the underlying asset, incorporating information from realized rates over the life of the market.

The result is an AMM that operates in the space of rates rather than in the space of prices. The state variable is not “how much is this token worth” but “what annualized implied rate is the

market pricing for this yield." This reproduces the language of fixed income in its purest form, exactly how a bond trader places orders by yield rather than by price.

Quadro 3 — Uniswap-style AMM and Pendle AMM compared

Uniswap-style AMM	**Pendle AMM**
Pool: ETH/USDC	Pool: PT/SY
Assets are independent	Assets are economically linked
Price divergence can be open-ended	PT converges to the underlying at maturity
Impermanent Loss comes from ETH/USDC relative price divergence	Liquidity Provider´s risk comes mainly from implied APY, maturity, liquidity and underlying risk
Price is the main signal	Price is translated into implied yield

Fonte: elaborada pela autora.

Quadro 4 — Components of the Pendle V2 pricing formula

Element	**What it is**	**Role in the curve**
$price_{a(t)}$	Price of the underlying asset denominated in PT (reciprocal of the PT price); ≥ 1 before maturity	The state variable of the AMM. Being the period interest factor, it converts directly into an implied rate
PT price	$PT\ price = \frac{1}{price_{a(t)}}$	Derived from the rate, not the other way around: the output, not the input
p	Proportion of PT in the pool	The only inventory input in the formula: it carries how much has been traded into the rate
Logit	$\ln\left(\frac{p}{1-p}\right)$	Sets the shape of the curve: flat at the centre, explosive at the extremes. Ordinary trades move the rate very little; pushing the rate to unrealistic levels requires disproportionate volume
rateScalar(t)	$rateScalar(t) = \frac{scalarRoot}{yearsToExpiry(t)}$	Compresses the curve as maturity approaches, the analogue of duration.

Element	What it is	Role in the curve
		Governs the sensitivity of the rate to trading
rateAnchor(t)	$rateAnchor(t) = price(t) - \frac{\ln\left(\frac{p}{1-p}\right)}{rateScalar(t)}$	Centres the curve on the current equilibrium rate of the underlying asset. Recalculated before each trade so that the rate does not drift without trading

Fonte: elaboração própria, com base em [17] e [29].

4.5.4 Dual Liquidity: AMM, Order Book, Router, and Flash Swap

Pendle operates two liquidity systems in parallel: the AMM and an order book. The swap router of the protocol seeks liquidity in both simultaneously, automatically computing the path that minimizes price impact and gas costs. A trader does not choose between them: the protocol decides the optimal combination.

The order book allows users to place limit orders for the purchase or sale of PT and YT at a specific implied APY. This is a critical detail of the interface: it is not a price in dollars, it is an implied rate. A trader does not say “I want to buy PT at $0.95.” A trader says “I want to buy PT at an implied APY of 8%.” This is the language of fixed income in the space of rates.

The interaction between the order book and the AMM is straightforward in principle. A limit order is executed when the implied APY of the AMM moves in the direction of the order. Any additional swap in that direction fills the order book before continuing to move the AMM. In practice, this means the order book deepens the AMM's liquidity at specific rate points. When the pool's implied APY reaches 8%, orders placed at 8% are executed first, and the AMM only continues to move if further trading pressure remains after those orders are filled.

The TradFi analogy is direct. In bond markets, market makers post bid and offer quotes at specific yields. Informed participants place orders at rates where they believe the market is mispriced, providing liquidity where the continuous market alone would be shallow. The Pendle order book replicates this function. Rather than requiring a human intermediary to match

liquidity at specific yields, the protocol maintains an on-chain order book where any participant can express a view at a chosen implied rate.

The flash swap mechanism extends the flexibility of the order book further. A taker order to buy YT can be matched with a limit order to buy PT, and vice versa. This may appear counterintuitive, but it follows from a mathematical equivalence built into Pendle's design: because PT and YT together reconstitute the underlying SY, buying YT is economically equivalent to selling PT against SY. The router recognizes this equivalence and matches the orders automatically. In traditional finance, this kind of matching would require a human intermediary or a specialized dealer with knowledge of the equivalence. In Pendle, it is executed by the protocol.

4.5.5 The Four-Layer Market Architecture

Most existing treatments of Pendle focus on the analogy that PT is described as similar to a bond, YT as similar to a coupon strip, and the analysis stops there. This analogy is accurate but superficial. It misses what is arguably the most significant contribution of the protocol.

In traditional finance, a functional fixed income market is not merely an instrument. It is an ecosystem with four components that must exist together for the market to work. The first is the instrument itself: what is being traded. The second is a pricing mechanism: how the price, or more accurately the rate, is formed. The third is order infrastructure: where and how orders are executed, and how liquidity is organized. The fourth is arbitrage: what keeps prices consistent across the different venues in which the instrument trades. If any of these components is absent, the market does not function well. A bond with no secondary market has no liquidity. A market with no efficient price formation produces no reliable rate signal. A system with no arbitrage produces inconsistent prices across venues.

Pendle replicates all four components on-chain. This is the core claim of this section, and the analytical argument developed throughout this dissertation.

Component 1: The Instrument. PT is the principal, YT is the yield stream, and the identity PT + YT = SY is the on-chain equivalent of the relationship principal + coupons = whole bond (Section 4.2). The institutional difference is that in traditional markets this separation is performed by specialized dealers, such as the primary dealers of the Treasury, while in Pendle any participant can perform the split and the protocol guarantees parity mathematically.

Component 2: Pricing by Yield. Bonds in traditional markets are priced by yield, not by price. A dealer does not say “I will buy at $98.50”. The dealer says “I will buy at 5.2%.” The price is derived from the rate, not the other way around. Pendle's AMM operates in the space of rates. When a participant places a limit order in the order book of Pendle, the order specifies an implied APY, not a dollar price. This is precisely the same logic. It is what distinguishes Pendle from what would otherwise be simply “a pool of tokens.” The geometry of the AMM, the logit curve, was constructed to operate in rates because that is what makes financial sense for this class of instruments.

Component 3: Order Infrastructure. Bond markets operate two liquidity layers, a continuous dealer market and an order book at specific rates, and Section 4.5.4 showed that Pendle reproduces exactly this pair: the AMM as the continuous layer, the order book by implied APY, with the router and the flash swap connecting them.

Component 4: Arbitrage Between the Two Systems. When the price of a bond in the dealer market diverges from the price in electronic orders, arbitrageurs buy at the cheaper venue and sell at the more expensive one until the prices converge. Pendle operates its own arbitrage bot that continuously aligns prices between the AMM and the order book. If the implied APY in the AMM is 8% but a limit order sits in the book at 7.8%, the bot corrects the discrepancy automatically. This matters because without arbitrage the two systems would give contradictory rate signals, which would destroy the reliability of the implied APY as market information.

The fundamental difference is the execution mechanism. In traditional finance, each layer involves human intermediaries operating with proprietary capital, subject to regulation, and enforceable through legal contracts. In Pendle, each layer is code: auditable smart contracts, configurable parameters, automated arbitrage. The economic result is similar. The institutional architecture is radically different. And that contrast, same economic function and entirely different infrastructure, is the argument this dissertation makes with the precision that the more common treatment of Pendle as “just a way to trade yield” does not attempt.

The layered structure of the Pendle protocol can be summarized as follows:

Layer 1: Yield Tokenization: SY → PT + YT, separating principal and yield.

Layer 2: Pricing: logit AMM operating in rate space, with fees proportional to yield moved rather than to principal.

Layer 3: Liquidity: the PT/SY pool provides continuous automatic liquidity; the order book provides peer-to-peer liquidity by implied APY; the flash swap connects PT and YT within a single pool; the router combines the AMM and the order book optimally.

Layer 4: Convergence: PT converges to one SY at maturity by construction; impermanent loss converges to zero at maturity by mathematical construction.

The on-chain implementation of this architecture is available for inspection in the public Pendle V2 repositories [33]. This is a point worth noting explicitly. In the traditional financial system, the pricing formulas used by a dealer are proprietary, and the audit trail of a market-making operation is available only to regulators and to the dealer itself. In Pendle, the pricing formulas are literally in the code, auditable by any participant. There is no intermediary between the financial theory and its execution: the contract is the market. This is a fundamentally different mode of institutional organization, and it is one of the properties that distinguishes crypto-native financial infrastructure from its off-chain counterpart.

5 RISK, MARKET STRUCTURE AND REGULATORY CHALLENGES

This chapter examines the main risks, structural limitations, and regulatory challenges associated with yield tokenization and the emergence of crypto-native fixed income-like markets. While previous chapters explored the economic and financial logic underlying protocols such as Pendle, this chapter focuses on the vulnerabilities and constraints that differentiate these markets from traditional fixed income systems.

Unlike sovereign or corporate bond markets, DeFi yield infrastructures operate through programmable and highly composable smart contract environments. As a result, the implied yields observed in tokenized yield markets incorporate not only time value and market expectations, but also multiple layers of technological, liquidity, collateral, governance, and protocol-specific risks.

The chapter closes with the regulatory dimension: Section 5.4 examines how three jurisdictions (the European Union, Dubai, and the United States) have approached the classification of instruments of this kind, and the tensions that yield tokenization creates for each of them.

5.1 Smart Contract and Protocol Risks

Section 3.4 established that DeFi yield-bearing positions typically derive from stacks of multiple protocols, transmitting composite yield and layered risks to Pendle positions specifically. This section focuses on one dimension of that risk structure that has particular weight for tokenized yield markets: the vulnerabilities embedded in the smart contracts and protocol infrastructures on which the entire stack depends.

In traditional financial systems, institutional intermediaries, legal frameworks, and centralized oversight often mitigate operational risks. By contrast, DeFi protocols rely primarily on programmable smart contracts executed autonomously on blockchain networks, often through immutable or semi-upgradable open-source code. Vulnerabilities in smart contract design may lead to bugs, exploits, unintended economic behavior, liquidity losses, or protocol failures. Even audited protocols remain exposed to implementation errors, governance attacks, oracle failures, and composability-related vulnerabilities [26].

It is worth noting that auditing in DeFi operates through a fundamentally different model than in traditional finance: rather than being conducted privately by a contracted firm with confidential access to internal records, smart contract audits are performed on publicly readable

code, and audit reports themselves are typically published in full. This model is complemented by bug bounty programs and continuous on-chain observation. The result is a structurally different architecture of trust. In traditional finance, trust is placed in institutions and the individuals who staff them: the auditing firm, the regulator, the custodian. In DeFi, trust is placed in publicly readable code and in the collective ability of many independent participants to verify it. Neither model eliminates the underlying risk, as recent exploit history in audited protocols demonstrates, but the locus of trust (institutions and people versus open code and distributed verification) is structurally different.

As a practical illustration of this open-verification model, the Pendle V2 codebase has been publicly audited by multiple independent firms, including WatchPug and Hickup, with the full audit reports openly available in the protocol's public repository [33]. Each report examines the code for implementation bugs, exploitable vulnerabilities, design flaws with potential economic consequences, and adherence to established security practices; findings are classified by severity (Critical, High, Medium, Low, Informational), each accompanied by a recommendation for remediation and a response from the protocol team indicating whether the issue was fixed, acknowledged, or disputed. Certain contracts, such as the marketV7 component of the Pendle AMM, were audited by more than one firm independently, a common practice in DeFi that reduces the probability of a single audit missing a vulnerability. The result is a verification process that is not only distributed across multiple auditors but also entirely public in its outputs, in contrast to the confidential audit reports typical of traditional financial infrastructure.

In the context of Pendle, these risks become particularly important because tokenized yield positions frequently depend on multiple external protocols simultaneously. A Pendle position may incorporate exposure not only to Pendle itself, but also to staking providers, restaking infrastructures, lending protocols, stablecoin issuers, oracle systems, bridges, and liquidity pools. This creates an interconnected dependency chain in which failures occurring in external infrastructures may directly affect the value, liquidity, pricing dynamics, or redemption assumptions of PTs and YTs, regardless of whether the Pendle smart contracts themselves function correctly.

The structural consequence of this dependency chain, that risk is reallocated rather than eliminated or amplified, was developed in Section 3.4 and returns in the discussion in Chapter 7. What matters at the level of smart contract risk specifically is the direction in which this reorganization shifts responsibility. On one hand, this reduces dependence on discretionary intermediary execution and makes financial logic publicly auditable on-chain. On the other

hand, it places greater responsibility on market participants themselves, who must understand the protocols, smart contract architectures, collateral structures, and dependencies embedded in the systems they interact with, rather than relying primarily on institutional protection or centralized oversight.

Consistent with the interpretation advanced in Section 3.4, the point is not to characterize DeFi as riskier or safer than traditional financial systems, but to observe that its risk structure operates through different mechanisms (code, protocol design, and open verification) and therefore requires different forms of analysis than those developed for institutionally intermediated systems.

An additional risk within this dependency structure concerns the underlying blockchain infrastructure itself. Pendle is deployed across multiple chains (including Ethereum, Arbitrum, BNB Chain, Optimism, Mantle, and Base), and the Standardized Yield interface was designed precisely to enable this abstraction, normalizing heterogeneous yield-bearing assets across different blockchain environments into a uniform representation that the tokenization layer can operate on. However, this multi-chain flexibility does not eliminate the structural concentration of tokenized yield markets in the Ethereum ecosystem, since the majority of the underlyings that make Pendle markets economically relevant (liquid staking tokens, liquid restaking tokens, yield-bearing stablecoins) remain anchored in Ethereum, whether directly on mainnet or on Layer 2s that inherit its security. As a result, tokenized yield markets carry a form of platform concentration risk at the level of the underlying economic activity, even when the protocol itself operates across multiple chains. The theoretical possibility of migration to alternative blockchains does not fully offset this dependency in the short and medium term, given the costs and time required for liquidity, tooling, and developer ecosystems to reorganize around a different base infrastructure.

5.2 Liquidity and Market Risks

Liquidity remains an important challenge for crypto-native fixed income-like markets. Although Pendle's specialized market microstructure, examined in detail in Section 4.5, was designed to improve execution efficiency and reduce fragmentation within tokenized yield markets, liquidity risk itself persists and affects PT and YT holders in specific ways.

Traditional bond markets generally benefit from institutional market makers, large issuance volumes, centralized settlement systems, and established benchmark curves that together produce deep, continuous liquidity across a wide range of maturities. By contrast, DeFi

yield markets remain relatively fragmented and dependent on protocol-specific liquidity conditions, which vary substantially across pools, underlyings, and time to maturity [16].

In Pendle, liquidity conditions directly affect the pricing efficiency of PTs and YTs. Lower liquidity may generate higher slippage on entry and exit, pricing distortions relative to the implied fair value, greater volatility in implied yields, and difficulties in exiting positions before maturity. These effects are particularly relevant for participants who intend to trade PTs or YTs before expiration rather than hold to maturity, since the ability to exit at an efficient price is not guaranteed by the redemption mechanism alone.

Despite the specialized architecture examined in Section 4.5, liquidity fragmentation and market depth limitations remain relevant challenges when compared to the scale, standardization, and institutional maturity of traditional sovereign and corporate bond markets. This limitation is one of the reasons why the implied yields observed in Pendle markets, as discussed in Section 4.3, reflect not only time value and market expectations but also market microstructure conditions specific to each individual market.

5.3 Depeg and Collateral Risks

A significant share of DeFi yield generation depends on stablecoins, collateralized assets, and interconnected blockchain-based infrastructures. For this reason, tokenized yield markets remain exposed not only to market volatility, but also to depeg risk, collateral stress, and disruptions affecting the mechanisms that support yield generation.

Stablecoins such as USDC, USDT, USDe, and other synthetic or yield-bearing assets may deviate from their expected reference value under stress conditions, including liquidity pressure, redemption imbalances, collateral deterioration, governance failures, or broader disruptions in crypto-financial infrastructure. Similarly, collateralized DeFi positions may be affected by changes in the quality, liquidity, or resilience of the assets and protocols supporting them.

In tokenized yield systems such as Pendle, these issues have specific consequences that connect directly to the analytical distinction developed in Section 4.2 between the **fixed-quantity property** and the **fixed-value property** of Principal Tokens. Because a PT redeems for a defined quantity of the underlying at maturity, a depeg or collateral impairment of the underlying does not affect the redemption quantity: the fixed-quantity property is preserved by construction. It does, however, affect the fixed-value property. If the underlying loses parity with its reference asset, the fiat value of the redemption changes correspondingly, and the

position no longer delivers a fixed return in fiat terms even though the underlying quantity remains unchanged. YT holders are exposed at a different point in the chain: since they receive the yield stream rather than the underlying, a depeg or collateral event that disrupts the yield-generating mechanism reduces the effective yield stream directly.

Under stress conditions, factors such as stablecoin depegging, collateral deterioration, validator slashing, staking instability, protocol insolvency, or failures in external infrastructure may affect market expectations regarding the underlying asset, and therefore influence PT and YT pricing before maturity. This should not be interpreted as evidence that tokenized yield instruments are inherently unstable. Traditional fixed income instruments also fluctuate before maturity when market expectations change. The relevant distinction is that, in DeFi, repricing may reflect not only yield and discounting conditions, but also protocol design, collateral architecture, validator performance, liquidity conditions, and the resilience of blockchain-based infrastructure [6].

At the same time, these risks should be understood within the context of a maturing ecosystem. As DeFi develops, improvements in protocol design, auditing standards, liquidity architecture, oracle systems, and market infrastructure tend to make tokenized yield structures progressively more resilient. The relevant analytical point is not that DeFi is exceptionally fragile by definition, but that it operates through a different and increasingly professionalized model of financial infrastructure, with its own forms of risk, transparency, and mitigation.

5.4 Regulatory Environment and Classification Challenges

Beyond the technical and market-structural risks examined above, yield tokenization also raises questions about how the resulting instruments should be classified within emerging regulatory frameworks. This section does not attempt a legal analysis; it identifies the classification tensions that yield tokenization creates for the regulatory categories currently in use.

Protocols such as Pendle introduce financial structures that economically resemble certain traditional instruments associated with fixed income and derivatives markets. Principal Tokens may exhibit characteristics functionally analogous to discounted fixed-income claims, while Yield Tokens provide tradable exposure to future yield streams. Despite these similarities, tokenized yield instruments are implemented as crypto-assets operating within decentralized smart contract infrastructures rather than through traditional securities issuance frameworks. This creates a classification question that has no clean answer under the categories currently

available to regulators: whether these instruments should be treated as securities, derivatives, structured financial products, tokenized claims, or virtual assets.

Within the European Union, the Markets in Crypto-Assets Regulation (MiCA) establishes a harmonized framework for crypto-assets and crypto-asset service providers. Nevertheless, the classification of DeFi yield tokenization mechanisms remains ambiguous under MiCA, precisely because these structures may partially resemble each of the categories the regulation covers without fitting any of them cleanly. At the same time, some DeFi systems may fall outside portions of traditional regulatory frameworks due to decentralization, absence of formal issuers, distributed governance structures, or autonomous smart contract execution, features that regulatory categories developed for centralized issuance do not naturally accommodate.

The Virtual Assets Regulatory Authority (VARA) framework in Dubai represents a different approach: rather than extending traditional financial classifications, it attempts to create dedicated regulatory categories specifically adapted to blockchain-based financial systems. The comparison between MiCA and VARA thus illustrates two possible strategies, extension versus new categorization, for handling instruments that do not fit existing frameworks.

The United States represents a third possible approach, distinct from both MiCA and VARA. Rather than establishing a unified crypto-asset framework, the U.S. regulatory environment remains fragmented across multiple agencies whose jurisdictional boundaries overlap. The Securities and Exchange Commission (SEC) applies the Howey Test to determine whether specific crypto-assets constitute investment contracts subject to securities regulation, while the Commodity Futures Trading Commission (CFTC) treats certain crypto-assets as commodities under its own jurisdiction. Individual states additionally impose specific requirements, such as New York's BitLicense. This fragmentation contrasts with the unified categorization approaches of MiCA and VARA and illustrates that the classification challenge posed by yield tokenization does not admit a single institutional solution across jurisdictions.

This dissertation does not attempt to provide a definitive legal classification of Pendle or similar protocols. The observation made here is different: yield tokenization creates instruments whose economic function is recognizable to fixed income analysis but whose institutional form does not correspond to any existing regulatory category. This tension is what makes the classification question analytically interesting, and it is unlikely to be resolved by simple mapping to pre-existing regulatory concepts.

The interaction between technological innovation, financial functionality, and regulatory adaptation is likely to become one of the defining issues shaping the future development of crypto-native capital markets. The framing developed throughout this chapter, that DeFi should not be understood as simply increasing financial risk but as reorganizing the institutional form through which financial risks and economic functions are structured, suggests that the same reorganization applies at the regulatory level: crypto-native instruments may require categorization frameworks whose logic reflects the technological substrate on which they operate, rather than being fitted into frameworks designed for a different institutional environment.

Quadro 2 — Regulatory Approaches to Yield Tokenization Across Jurisdictions

Dimension	MiCA (European Union)	VARA (Dubai, UAE)	United States
Regulatory strategy	Unified extension of existing categories through a harmonized framework applicable across all member states.	Dedicated new categorization designed specifically for blockchain-based financial systems.	Fragmented multi-agency approach without a unified crypto-asset framework, with jurisdictional overlap between federal regulators and additional state-level rules.
Principal legal instrument	Regulation (EU) 2023/1114 on Markets in Crypto-Assets (MiCA).	VARA Rulebooks and Virtual Assets and Related Activities Regulations 2023.	Securities Act (Howey Test) applied by the SEC; Commodity Exchange Act applied by the CFTC; state-level frameworks such as New York's BitLicense.
Geographical scope	European Union (harmonized across 27 member states).	Emirate of Dubai (UAE, sub-national jurisdiction).	United States (federal and state).
Classification of yield tokenization instruments	Ambiguous — the structures may partially resemble each of the categories the regulation covers without fitting any of them cleanly.	More adaptable — the categories were built with blockchain-based instruments in mind, allowing more natural accommodation of tokenized yield structures.	Case-by-case — classification depends on which agency evaluates the instrument and how existing tests (Howey, commodity definitions) are applied.

Source: prepared by the author, drawing on Regulation (EU) 2023/1114 (MiCA); VARA Rulebooks; U.S. Securities and Exchange Commission and Commodity Futures Trading Commission regulatory frameworks; and the analytical framework developed in this dissertation.

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GLOSSARY

This glossary is organized into three sections. Section 1 lists the blockchain networks and DeFi protocols referenced in the dissertation.

Section 2 lists the traditional finance and economics concepts invoked in the analysis.

Section 3 lists the DeFi and Pendle-specific technical vocabulary. Entries are alphabetically ordered within each section per FGV norma; each entry consists of the term in bold uppercase followed by a one-sentence definition. To be finalized once Chapters 6 through 8 are written and the term set is stable.

1. Blockchain Networks and Protocols

AAVE — A decentralized lending protocol operating pool-based liquidity markets in which lenders receive interest-bearing aTokens against their deposits and borrowers post overcollateralized positions, with supply and borrow rates determined algorithmically by the pool's utilization ratio.

ARBITRUM — An Ethereum Layer 2 network based on optimistic rollup technology, providing lower fees and higher throughput while inheriting Ethereum's security; one of the blockchains on which Pendle is deployed.

BASE — An Ethereum Layer 2 network developed by Coinbase, based on the OP Stack; one of the blockchains on which Pendle is deployed.

BITCOIN — The first decentralized cryptocurrency, introduced in 2008, designed to enable peer-to-peer electronic transactions without a trusted third party.

BNB CHAIN — A blockchain network developed within the Binance ecosystem, operating as an alternative Layer 1 environment for DeFi applications; one of the blockchains on which Pendle is deployed.

COMPOUND — A decentralized lending protocol structurally similar to Aave, in which lenders receive cTokens against their deposits and interest rates are set algorithmically by pool utilization.

CURVE FINANCE — A decentralized exchange specialized in trading between assets expected to trade near parity, most notably stablecoins and pegged assets such as stETH against ETH, through an AMM curve designed to concentrate liquidity around the peg and reduce slippage for parity trades.

EIGENLAYER — A protocol introducing restaking on Ethereum, in which staked ETH or liquid staking tokens can be committed to secure additional services beyond consensus.

ETHENA — A DeFi protocol issuing USDe, a synthetic dollar backed by delta-hedged crypto collateral rather than by fiat reserves; represents an alternative model of yield-bearing stablecoin design.

ETHER.FI — A liquid restaking protocol issuing a liquid restaking token representing exposure to restaked ETH and the additional rewards of actively validated services the underlying stake secures.

ETHEREUM — A public blockchain supporting general-purpose smart contracts, on which the majority of DeFi activity takes place.

LIDO — The leading liquid staking protocol on Ethereum, issuing stETH as a tokenized claim on staked ETH that continues to accrue consensus rewards while remaining transferable and composable with other DeFi protocols.

MAKERDAO — A decentralized protocol for the issuance of collateralized stablecoins, historically the issuer of the DAI stablecoin backed by on-chain collateral positions.

MANTLE — An Ethereum Layer 2 network based on modular rollup architecture; one of the blockchains on which Pendle is deployed.

MORPHO — A decentralized lending protocol that layers a peer-to-peer matching mechanism on top of pool-based lending markets to improve capital efficiency for both lenders and borrowers.

NOTIONAL FINANCE — The earlier DeFi protocol focused on fixed-rate on-chain lending through instruments called fCash, whose AMM design serves as the direct foundation for the Pendle V2 AMM.

OPTIMISM — An Ethereum Layer 2 network based on optimistic rollup technology, whose OP Stack technology also underlies other Layer 2 networks such as Base; one of the blockchains on which Pendle is deployed.

PENDLE — A DeFi protocol enabling the tokenization of future yield through the separation of yield-bearing assets into Principal Tokens and Yield Tokens, examined as the empirical case study of this dissertation.

RENZO — A liquid restaking protocol operating on EigenLayer, issuing a liquid restaking token representing exposure to restaked ETH and layered AVS rewards.

ROCKET POOL — A decentralized liquid staking protocol on Ethereum, issuing rETH as a tokenized claim on staked ETH, distinguished from Lido by its permissionless validator set.

UNISWAP — The first major decentralized exchange based on automated market making, whose constant-product design (V2) and later concentrated-liquidity design (V3) shaped the standard vocabulary of on-chain trading.

YEARN FINANCE — A DeFi protocol offering automated vault strategies that aggregate multiple yield-generating operations across other protocols into single deposit interfaces.

2. Traditional Finance and Economics Terms

BONDLESS MARKET — A market that generates prices from which implied interest rates can be inferred but which lacks the sovereign or corporate bond instruments from which such rates are conventionally extracted; the term applies to DeFi and motivates the alternative rate-inference methodologies discussed in the yield tokenization literature.

BOOTSTRAPPING — In traditional fixed income, the sequential procedure used to extract a spot rate curve from the observable prices of coupon-bearing bonds. Starting from the shortest-maturity zero-coupon instrument, whose yield is directly a spot rate, the spot rate at each subsequent maturity is derived algebraically by using the spot rates already known to discount the intermediate coupon payments of a longer bond, leaving the spot rate at the bond's maturity as the only unknown to solve for. Repeating this step across maturities produces the full spot rate curve. Standard practice in central banks, fixed income desks, and risk management systems; the zero-coupon Treasury curve published on market data terminals is constructed this way.

CONVEXITY — The second-order measure of a fixed income instrument's price sensitivity to yield changes; captures the curvature of the price-yield relationship not accounted for by duration alone.

COUPON STRIP — A tradable claim on the coupon component of a decomposed bond, separated from the principal component by primary dealers in traditional fixed income markets and invoked in this work as the closest traditional analogue to a Yield Token.

COUPON-BEARING BOND — A fixed income instrument that makes periodic interest payments (coupons) during its life in addition to returning the principal at maturity, in contrast to a zero-coupon bond which pays only a single amount at maturity.

DEPOSITORY TRUST COMPANY (DTC) — The United States central securities depository that standardizes the custody, transfer, and settlement of securities issued by heterogeneous entities, invoked in this work as an analogy for the standardization role of Standardized Yield in Pendle.

DISCOUNT RATE — The rate used to translate a future cash flow into its present value.

DURATION — A measure of a fixed income instrument's price sensitivity to changes in yield, calculated as a weighted average of the times at which its cash flows are received.

EUROCLEAR — A European central securities depository providing settlement and custody services across multiple jurisdictions, invoked in this work as an analogy for the standardization role of Standardized Yield in Pendle.

FACE VALUE — The principal amount promised by a fixed income instrument at maturity, used as the reference against which the discount at issuance and the yield to maturity are calculated.

FIXED-YIELD EXPOSURE — The economic position obtained by holding a Principal Token to maturity, in which the return realized between purchase and redemption is determined at the time of purchase; the exposure is fixed in units of the underlying and, when the underlying is stablecoin-denominated, also fixed in fiat terms.

INTEREST RATE RISK — The risk that changes in market interest rates will affect the price of a fixed income instrument or the value of a portfolio holding such instruments, distinct from credit risk.

INTERNALIZATION OF LOSSES — The mechanism by which losses arising from failed structures are absorbed directly by the participants who assumed the underlying risk, without institutional backstop; characteristic of decentralized systems in contrast with the socialization of losses observed in traditional financial systems.

MATURITY — The date at which a fixed income instrument's principal becomes due; in Pendle, the date at which a Principal Token redeems for the underlying.

MONEY MULTIPLICATION (AS ANALOGY) — The economic mechanism by which an initial base of assets or collateral supports multiple layers of financial claims through the traditional banking system, invoked in this work as a partial analogy for DeFi composability while acknowledging that the two mechanisms operate through fundamentally different institutional arrangements.

MORAL HAZARD — The distortion of incentives that arises when the party bearing the cost of a failure is different from the party whose decisions influence the probability of failure; central to the comparison between traditional financial systems, where institutional support may generate this distortion, and decentralized systems, where failures are more directly internalized by participants.

PRESENT VALUE (PV) — The value today of a future cash flow, obtained by discounting the future amount at an appropriate rate over the intervening horizon.

QUADRO — In ABNT terminology, a categorical or descriptive comparison presented in tabular form, distinguished from Tabela, which is reserved for quantitative data.

REDEMPTION VALUE — The amount of the underlying asset delivered to the holder of a Principal Token at maturity; fixed in quantity for all Pendle markets, and fixed in fiat value only when the underlying is stablecoin-denominated.

RISK-FREE RATE — The theoretical return on an investment carrying no risk of default and no interest rate risk over the holding period; in practice, approximated by short-term sovereign debt of governments considered least likely to default.

SECURITIZATION (AS ANALOGY) — The traditional finance mechanism in which future cash flows from underlying assets are pooled, structured, and transformed into tradable securities; invoked in this work as an analogy for yield tokenization while noting that the two

constructions differ substantially in institutional apparatus — there is no special purpose vehicle, no legal assignment, no tranches, and no rating in the Pendle case.

SOCIALIZATION OF LOSSES — The mechanism by which losses arising from institutional failures are absorbed by parties who were not participants in the failed structure, typically through public intervention, monetary expansion, or the balance sheets of third parties; contrasted in this work with the internalization of losses that characterizes decentralized systems.

SPECIAL PURPOSE VEHICLE (SPV) — A legal entity created in traditional securitization structures to isolate the pooled assets from the originator's balance sheet, providing the legal apparatus that gives structured claims their institutional standing; absent by construction in DeFi yield tokenization.

SPOT RATE — The interest rate applicable to a single cash flow at a given horizon, extracted from the price of a zero-coupon instrument of that maturity.

TERM STRUCTURE OF INTEREST RATES — The set of spot rates applicable to different horizons under prevailing market conditions, typically read from a yield curve constructed from zero-coupon instruments.

TIME VALUE OF MONEY — The principle that a unit of currency available today is worth more than the same unit received at a future date, because the present unit can be invested and earn a return over the intervening period.

TRADFI — Traditional finance; the established financial system of banks, exchanges, custodians, and regulated institutions operating off-chain.

TRANCHE — In traditional securitization, one of several classes of claims on a pool of underlying assets, ordered by seniority of payment and typically rated separately; absent in DeFi yield tokenization, where PT and YT are not subordinated to one another.

YIELD CURVE — The graphical representation of the relationship between the yields of fixed income instruments and their times to maturity, used to display the term structure of interest rates.

YIELD TO MATURITY (YTM) — The discount rate that equates the present value of a bond's future cash flows with its current market price.

ZERO-COUPON BOND — A fixed income instrument that promises a single payment at maturity and pays no intermediate coupons, purchased at a discount to face value.

3. DeFi and Pendle-Specific Technical Terms

ACTIVELY VALIDATED SERVICE (AVS) — In the EigenLayer framework, an additional protocol that opts into restaked security in exchange for paying rewards to restakers.

ARBITRAGE (IN PENDLE) — The automated realignment of implied APY between the AMM and the order book, executed by a protocol-level bot that trades against price discrepancies until the two venues converge.

ATOKEN — Interest-bearing token issued by Aave to lenders, representing a claim on a share of a lending pool whose value appreciates continuously as borrower interest accrues.

AUTOMATED MARKET MAKER (AMM) — A decentralized exchange design in which trade pricing is determined by a mathematical function of the pool's inventory rather than by matched buy and sell orders.

BEHAVIOR OF YIELD STREAM — The realized trajectory of yield produced by a yield-bearing asset over the life of a Pendle market; the primary source of exposure for a Yield Token holder, distinct from the redemption exposure that characterizes Principal Tokens.

BLOCKCHAIN — A distributed ledger in which transactions are grouped into cryptographically linked blocks maintained by a network of independent nodes.

BRIDGE (CROSS-CHAIN BRIDGE) — An infrastructure component that enables the transfer of tokens or messages between distinct blockchain networks; a common point of vulnerability in DeFi due to the concentration of value locked in bridge contracts and the complexity of cross-chain security models.

BUG — In DeFi, an error in the code of a smart contract or protocol where the implemented logic diverges from the intended specification, producing unintended technical or economic outcomes; distinct from an exploit in that no external actor is required, and distinct from unintended economic behavior in that the code itself is doing something different from what the developers meant to program.

COLLATERAL IMPAIRMENT — A deterioration in the quality, value, or liquidity of the assets supporting a protocol or stablecoin structure, which may reduce the reliability of the redemption mechanism and affect the pricing of instruments dependent on that collateral.

COMPOSABILITY — The property by which the output of one DeFi protocol can be used directly as input to another, allowing protocols to be combined into layered financial structures.

COMPOSITE IMPLIED YIELD — An implied yield inferred from Principal Token pricing that reflects the market's expectation over an entire stack of dependencies embedded in the underlying asset, rather than the yield of a single protocol.

CONSTANT PRODUCT AMM — The AMM design characteristic of Uniswap V2 and its derivatives, in which the invariant $x \cdot y = k$ governs pricing across the two-asset pool.

CONVERGENCE PROPERTY (OF PT) — The property by which the price of a Principal Token converges to the redemption value of one unit of the underlying at the maturity date, regardless of the price path taken to reach expiration.

CRYPTO-NATIVE FIXED INCOME — Fixed income structures whose underlying cash flows originate within decentralized protocols and whose market infrastructure is constructed on-chain, in contrast to Real World Asset tokenization in which off-chain fixed income instruments are wrapped as on-chain tokens.

CTOKEN — Interest-bearing token issued by Compound to lenders, analogous in function to Aave's aToken.

CYPHERPUNK — A movement of cryptographers and software developers, dating to the early 1990s, focused on the use of cryptography and open-source software to protect privacy and reduce reliance on centralized intermediaries.

DECENTRALIZED FINANCE (DEFI) — A set of blockchain-based financial protocols providing trading, lending, borrowing, liquidity provision, and related services through smart contracts, without traditional intermediaries.

DEPEG — The event in which a pegged asset, typically a stablecoin or a liquid staking derivative, deviates materially from its reference value.

EIP _ Ethereum Improvement Proposal.

ERC (Ethereum Request for Comments) — A technical standard that specifies an interface, the set of functions and events, that a smart contract must implement to interoperate with other contracts and applications on the Ethereum blockchain. Standards emerge through community proposal and adoption rather than through a central authority.

ERC-20 — The Ethereum token standard defining a common interface for fungible tokens, which underlies most tradable DeFi assets. Interface for fungible tokens. Defines six functions (totalSupply, balanceOf, transfer, transferFrom, approve, allowance) and two events (Transfer, Approval). Each token has its own contract maintaining an address => uint256 mapping of balances. It does not embed yield or maturity logic — it is the minimum standard for a transferable unit.

ERC-4626 — Interface for tokenized yield-bearing vaults. Extends ERC-20 (the vault share itself is an ERC-20 token) and adds specific functions: deposit, mint, withdraw, redeem, convertToShares, convertToAssets, totalAssets. Standardizes how a contract that receives an underlying asset and issues shares (proportional to the vault's value) must behave. Solves the problem that previously each yield protocol had its own interface, forcing case-by-case integrations.

ERC-5115 — Standardized Yield (SY) standard designed by Pendle. Extends ERC-20 and defines a unified interface to wrap any yield-bearing asset (stETH, aUSDC, cDAI, LP tokens, etc.) into a canonical form. Main functions: deposit (accepts multiple input tokens), redeem, exchangeRate (returns the conversion rate between SY and the underlying asset), getRewardTokens, claimRewards. Unlike ERC-4626, which assumes a single underlying asset, ERC-5115 supports multiple input/output tokens and multiple reward tokens — necessary because Pendle must tokenize yield from heterogeneous sources within a single interface before splitting into PT and YT.

ETHEREUM VIRTUAL MACHINE (EVM) — The execution environment in which Ethereum smart contracts run, replicated across all validating nodes of the network.

EXPLOIT — In DeFi, the act of leveraging a vulnerability in a smart contract or protocol design to extract value or produce economic outcomes contrary to the protocol's intended behavior; distinct from a bug in that it involves an intentional actor, and typically resulting in loss of user funds.

FCASH — The fixed-rate lending instrument developed by Notional Finance, structurally analogous to a zero-coupon bond and conceptual predecessor to the Pendle V2 AMM design.

FIXED-QUANTITY PROPERTY — The property by which a Principal Token redeems for a defined quantity of the underlying asset at maturity regardless of the price path; holds for all Principal Token markets.

FIXED-VALUE PROPERTY — The property by which the fiat value of a Principal Token's redemption amount is predictable at the time of purchase; holds only when the underlying is denominated in a stable unit of account such as a high-quality stablecoin.

FLASH SWAP — A Pendle mechanism that leverages the mathematical equivalence between PT, YT, and SY to match orders across these components within a single atomic transaction, allowing a YT purchase to be filled by a PT purchase and vice versa without requiring a separate counterparty for each token.

FOUR-LAYER MARKET ARCHITECTURE — The analytical framework developed in this dissertation that identifies the four components required for a functioning fixed income market (instrument, pricing by yield, order infrastructure, and arbitrage) and observes that Pendle reproduces all four on-chain through different mechanisms than those employed in traditional finance.

GOVERNANCE ATTACK — An exploit in which a participant or coalition accumulates enough governance tokens to pass proposals that extract value from the protocol or alter its parameters in ways contrary to the interests of ordinary users.

GOVERNANCE TOKEN — A token issued by a DeFi protocol that confers voting rights over protocol parameters, upgrades, and treasury allocations, often distributed as a yield component within incentive-driven strategies.

HICKUP HH3 — Smart contract security auditing firm (part of the Halborn security ecosystem). Author of one of the public audit reports on the Pendle V2 codebase. [GitHubPendle_Audits]

HICKUP HH3 — Smart contract security audit report on the Pendle V2 marketV7 contract (the AMM component), authored by Hickup. Available in the audits folder of the Pendle public repository. https://github.com/pendle-finance/pendle-core-v2-public/tree/main/audits

IMPERMANENT LOSS — The loss experienced by an AMM liquidity provider when the external market price of the paired assets moves away from the entry ratio, realized whenever the LP exits at a different price ratio.

IMPLIED APY — The annualized rate implicit in Pendle market pricing, obtained from the relationship between the current Principal Token price, its redemption value, and the remaining time to maturity; the unit in which limit orders are placed on the Pendle order book.

IMPLIED YIELD / IMPLIED RATE — A yield inferred from the market price of a tradable instrument rather than directly quoted; in Pendle, the yield inferred from the pricing of a Principal Token against its underlying.

LAYER 1 (L1) — A base-layer blockchain network that operates its own consensus mechanism independently, without relying on another blockchain for security; examples include Ethereum, Bitcoin, Solana, and BNB Chain. The term describes an architectural position (base layer versus higher layers), not an environment type (like mainnet and one or more testnets).

LAYER 2 (L2) — A blockchain network that operates on top of a base layer (most commonly Ethereum) to provide lower fees and higher throughput while inheriting the security of the base layer; examples include Arbitrum, Optimism, and Base.

LAYERED YIELD — A yield-bearing position whose total return derives from multiple stacked protocols, each contributing an additional layer of yield and an additional set of dependencies.

LIQUID RESTAKING TOKEN (LRT) — A tokenized claim on a restaked position, representing exposure to base staking yield plus the additional rewards of any actively validated services the underlying stake secures.

LIQUID STAKING DERIVATIVE / TOKEN (LSD / LST) — A tokenized claim on a staked position that continues to accrue staking rewards while remaining freely transferable and composable with other DeFi protocols; examples include Lido's stETH and Rocket Pool's rETH.

LIQUIDATION — The forced closing of a collateralized position when the value of the collateral falls below a protocol-defined threshold.

LIQUIDITY PROVIDER (LP) — A participant who deposits assets into an AMM pool in exchange for a claim token entitling them to a share of trading fees and pool inventory.

LOGIT FUNCTION — The function ln(p/(1−p)), inverse of the sigmoid, used in the Pendle V2 AMM to produce a pricing curve that is flat near equilibrium and explosive at the extremes, reproducing in code the behavior expected of a well-functioning fixed income market.

LP TOKEN — A tokenized claim representing a liquidity provider's share of an AMM pool.

MAINNET — The primary production environment of a blockchain network, on which real-value transactions are settled; distinguished from testnets, which operate as parallel networks used for development and experimentation without economic consequence. The term describes a type of environment (production versus test), not an architectural position: both Layer 1 networks (e.g., Ethereum, Solana) and Layer 2 networks (e.g., Arbitrum One) maintain their own mainnets.

MEV (MAXIMAL EXTRACTABLE VALUE) — Value that can be extracted from block production by reordering, inserting, or censoring transactions within a block.

MICA — Markets in Crypto-Assets Regulation, the European Union framework for the regulation of crypto-asset markets.

OFF-CHAIN — Referring to activity, assets, or infrastructure that exists outside of a blockchain, typically within traditional financial or institutional systems.

ON-CHAIN — Referring to activity, assets, or infrastructure that exists directly on a blockchain, executed and settled through its native protocol layer.

ORACLE — An external data feed that provides off-chain information, typically prices, to on-chain smart contracts.

ORDER BOOK (PENDLE) — The on-chain component of the Pendle liquidity engine that accepts limit orders specified by implied APY rather than by dollar price, complementing the AMM as a second layer of liquidity available to the same market.

PEGGED ASSET — A token designed to track the value of a reference asset; examples include stablecoins pegged to fiat currencies and liquid staking derivatives pegged to their underlying.

PENDLE MARKET — A smart contract deployed by the MarketFactory contract that tokenizes a specific underlying asset with a specific expiry date. Each market enables trading between PT and its corresponding SY, holds its own liquidity pool, and has its own built-in geometric-mean oracle. YT is not traded directly but remains tradable through flashswap algorithms. The coexistence of multiple markets on the same underlying, each with a different expiry, forms the Pendle term structure of interest rates for that underlying. https://docs.pendle.finance/pendle-v2-dev/HighLevelArchitecture

PLATFORM CONCENTRATION RISK — The risk arising from the fact that a large share of DeFi activity, including the underlyings that make tokenization protocols such as Pendle economically relevant, remains anchored in a single blockchain ecosystem; multi-chain deployment of a protocol reduces this risk at the interface level but does not eliminate it at the level of underlying economic activity.

PRINCIPAL TOKEN (PT) — The Pendle token representing a claim on the principal component of a yield-bearing asset, redeemable for a defined amount of the underlying at maturity; functionally analogous to a zero-coupon bond in the strict sense when the underlying is stablecoin-denominated.

PROOF-OF-STAKE (POS) — A consensus mechanism in which validators are selected to produce blocks in proportion to the quantity of native tokens they have committed as stake.

PROTOCOL INSOLVENCY — A situation in which a DeFi protocol becomes unable to sustain redemptions, honor its liabilities, or maintain the economic incentives required for its continued operation.

RATE SPACE — The pricing dimension in which the state variable is an annualized rate rather than a dollar price; the space in which the Pendle V2 AMM operates, in contrast to general-purpose AMMs that operate in price space.

RATEANCHOR — A Pendle AMM parameter that shifts the pricing curve vertically to center liquidity around the current equilibrium implied rate for the underlying asset, incorporating information from realized rates over the life of the market.

RATESCALAR — A Pendle AMM parameter defined as scalarRoot divided by the years remaining to expiry, which compresses the pricing curve as maturity approaches and effectively serves as a duration analogue for the market.

REAL WORLD ASSET (RWA) TOKENIZATION — The tokenization of off-chain financial instruments (Treasury bills, money market fund shares, private credit) as on-chain tokens, in contrast to crypto-native fixed income structures whose cash flows originate on-chain.

RESTAKING — The commitment of already-staked capital, or its liquid-staked representation, to secure additional services beyond the underlying consensus protocol.

ROUTER (PENDLE) — The Pendle protocol component that combines the AMM and the order book automatically for each trade, computing the execution path that minimizes price impact and transaction costs.

SCALARROOT — The base parameter from which rateScalar is computed; a market-specific constant chosen at pool creation that determines how aggressively the pricing curve compresses as maturity approaches.

SLASHING — A penalty applied by a proof-of-stake protocol to validators that violate consensus rules, resulting in the loss of part or all of their stake.

SLIPPAGE — The difference between the expected execution price of a trade and the price actually realized, arising from limited liquidity depth or from price movement during execution; a central concern in AMM-based markets, including Pendle's.

SMART CONTRACT — A program deployed on a blockchain whose logic is executed automatically by the network according to its coded rules.

SMART CONTRACT RISK — The risk of financial loss arising from bugs, vulnerabilities, or exploits in the code of a smart contract.

STABLECOIN — A token designed to maintain a stable value against a reference asset, most commonly a fiat currency such as the U.S. dollar.

STAKING — The commitment of native tokens to a proof-of-stake protocol in exchange for consensus rewards funded by new issuance and transaction fees.

STAKING INSTABILITY — Unexpected variations in staking rewards, validator performance, withdrawal conditions, or staking participation dynamics that affect the yield generated by staked positions.

STANDARDIZED YIELD (SY) — A Pendle token standard (ERC-5115) that wraps a yield-bearing asset into a normalized interface suitable for splitting into Principal and Yield Tokens; functions analogously to a securities depository by abstracting the differences between heterogeneous yield sources into a single interface.

STETH — The liquid staking token issued by Lido, representing a claim on staked ETH that accrues Ethereum consensus rewards while remaining transferable.

TOKENIZATION — The process of creating a transferable and programmable representation, on a blockchain, of an economic right associated with an underlying asset. **Technically**, tokenization consists of deploying a smart contract that implements a token standard (such as ERC-20) and maintains an on-chain ledger of balances associating token units with specific addresses; the token itself is a record of claim entries governed by the contract's rules, not the underlying asset. **Economically,** tokenization defines what right the token represents — ownership, credit, yield, or another claim — and how that right relates to the underlying asset, which may exist off-chain (real estate, fiat currency) or on-chain (native crypto-assets, yield-bearing positions). The technical construction determines how the token behaves as an instrument; the economic design determines what the token means as a claim.

TOTAL VALUE LOCKED (TVL) — The aggregate value of assets deposited in a DeFi protocol at a given point in time, commonly used as a measure of protocol scale.

UNDERLYING — The yield-bearing asset that is deposited into a Pendle SY contract (for example, stETH, aUSDC, cDAI). It is generated by another DeFi protocol and exists independently of Pendle.

UNDERLYING STACK — The underlying together with all the protocols and mechanisms it depends on (Lido for stETH, Aave for aUSDC, plus any oracles, bridges, or staking infrastructure involved). The stack determines both the yield and the layered risks that reach PT and YT holders.

UNINTENDED ECONOMIC BEHAVIOR — In DeFi, a situation in which a smart contract executes exactly the logic it was programmed with but produces an economic outcome different from the one the developers anticipated; distinct from a bug in that the code performs its intended function, and distinct from an exploit in that no external malicious actor is required.

USDC — A U.S. dollar-backed stablecoin issued by Circle, redeemable one-to-one for fiat and backed by reserves held with regulated financial institutions; one of the reference stablecoins in DeFi.

USDE — A synthetic dollar issued by Ethena, backed by delta-hedged crypto collateral rather than by fiat reserves; represents an alternative model of yield-bearing stablecoin whose stability depends on the continued functioning of its hedging mechanism.

USDT — A U.S. dollar-backed stablecoin issued by Tether, redeemable one-to-one for fiat and backed by a reserve portfolio; the largest stablecoin by market capitalization and one of the reference stablecoins in DeFi.

UTILIZATION RATIO — In pool-based lending, the ratio of borrowed assets to total assets in the pool, used by the protocol's interest rate curve to set supply and borrow rates.

VARA — The Virtual Assets Regulatory Authority of Dubai, one of the emerging jurisdiction-specific frameworks for the regulation of crypto-asset activities.

VAULT STRATEGY — An automated DeFi construct that aggregates multiple yield-generating operations into a single deposit interface, redistributing user capital across component protocols according to a predefined algorithm; examples include the strategies offered by Yearn Finance.

WATCHPUG — Asian smart contract security auditing firm specializing in DeFi protocols. Author of one of the public audit reports on the Pendle V2 codebase.[GitHubPendle_Audits]

YIELD FARMING — The practice of moving capital across DeFi protocols to capture the highest available yields, often augmented by governance token incentives.

YIELD TOKEN (YT) — The Pendle token representing a claim on the future yield stream generated by a yield-bearing asset until maturity, complementary to the Principal Token.

YIELD TOKENIZATION — The decomposition of a yield-bearing asset into separately tradable claims on its principal and its future yield, allowing each to be priced and held independently.

YIELD TOKENIZATION — The specific application of tokenization in which a yield-bearing asset is decomposed into two separately tradable claims, one representing the principal and one representing the future yield stream. **Technically,** this is implemented by depositing

the underlying yield-bearing asset into a smart contract that mints two new tokens — the Principal Token and the Yield Token — each governed by its own contract and redemption rules. **Economically,** this creates two derivative instruments whose combined economic content is equivalent to holding the original position, but whose independent tradability allows each component to be priced, held, and used as collateral separately.

YIELD-BEARING ASSET — A token whose underlying position generates yield over time, through lending interest, staking rewards, trading fees, or a composite of these, accruing to the token holder.

YIELDSPACE — An earlier AMM design that introduced time as an exponent in the pricing function to accommodate instruments with a fixed maturity, but which lacks free parameters to specialize the curve to the yield profile of the specific underlying asset, a limitation that motivated the Pendle V2 adoption of the Notional Finance model.